

CONDUCTOR BOUNDARIES AND PRIMARY STRUCTURES IN MULTIPLICATION FAILURE

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ABSTRACT. We study multiplication failure through its full coherent cokernel, including the primary structure on the conductor boundary. For every complex local algebra with Hilbert function $(1, 2, 2)$ we classify the complete failure scheme of three generators. The factors of the quadratic relation determine its embedded surfaces and a possible embedded point at the extreme-corank locus. The reduced support is a normal determinant divisor; its generic multiplicity is two, while the full scheme has nilradical index four. The proof uses a factorization of the full Fitting ideal valid for every cube-zero augmentation ideal. A conductor-kernel formula locates the associated strata and their action-rank jumps, and an algebra deformation relates the two primary types. More generally, products of a unital codimension- r subbundle of a finite locally free algebra over a $\mathbb{Q}$ -scheme stabilize in degree $r + 1$. We prove that this bound is sharp in every codimension and generating rank. The codimension-two presentation, its proper quotient-algebra incidence, and a connected higher-contact family with unbounded nilpotent complexity complement the boundary classification. A regularity-controlled conductor theorem identifies the finite and global multiplication cokernels for zero-dimensional projective schemes, including finite flat families and arbitrary base change.

1. INTRODUCTION

The multiplication map of a linear series can fail to be surjective with a scheme structure which changes even when its support does not. We study the actual multiplication cokernel on generating Grassmannians of finite algebras, and transport it to global sections of projective schemes. The central link is a codimension bound over the coefficient ring, followed by a conductor incidence which retains the local rings of the failure scheme.

Let $W \subset B$ contain 1 and have locally free quotient of rank r . Theorem 4.1 proves $W^{r+1} = \mathcal{O}[W]$ over every $\mathbb{Q}$ -algebra. The proof compresses the commuting multiplication operators to B/W . Their commutators land in the quadratic residual image; polarized Cayley–Hamilton then bounds the residual filtration without discarding nilpotents on the base. In codimension two the entire stable failure scheme is therefore defined by two by two minors of one degree-three residual matrix (Theorem 7.1). Its entry ideal defines a different scheme: the extreme-corank subalgebra locus.

Off that locus the full failure scheme is an open relative Grassmannian over $\text{Hilb}^2(B)$. This is an isomorphism of schemes, not only a parameterization of geometric points. The proper incidence extending it has fibres controlled by the conductor of the codimension-two subalgebra. When the conductor action has rank one, the fibre is the length-two quotient scheme of a rank-three algebra, expressed by a binary cubic. When the action has rank two, it is the spectrum of a quadratic algebra (Proposition 7.2). The rank-three and rank-four alternatives of Theorem 6.4 thus govern one incidence over the full multiplication failure scheme; they do not by themselves classify every primary component on its extreme-corank locus.

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The incidence yields a systematic non-Cartier family. For $B_h = \mathbb{C}[x, y]/(x^2, xy, y^h)$, $h \geq 5$, the full failure scheme of unital $(h-1)$ -planes has local rings which are localized polynomial extensions of

$$\mathbb{C}[b, u, v]/(F_h, vF_{h-1}, bv, b^2u), \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1}.$$

Their exact primary decomposition is

$$(u, v) \cap (b^2, bv, F_h, vF_{h-1}).$$

The minimal support has codimension $h-3$, an embedded component lies over $b = u = v = 0$, and the nilradical has exact index $2h-3$ (Theorem 9.1). The mixed equations bv and b^2u couple the extra tangent direction to the higher contact coefficients. Corollary 9.2 also supplies genuine three-generator examples. The unit-cover description places these local rings on smooth covers of the full generating Grassmannians, rather than on selected linear slices.

A general conductor theorem completes the passage to global geometry. If I_Z is the saturated ideal of any zero-dimensional scheme in $\mathbb{P}^e$, then $n \geq 2 \operatorname{reg} I_Z - 1$ suffices to identify the global and finite multiplication cokernels in every degree $m \geq 2$ (Theorems 10.1 and 10.2). It applies to nonmonomial local schemes, unions, and finite flat families with a uniform regularity bound. The proof fills the conormal layer and the ordinary ideal-square layer. The square-regularity bound is the classical input of [35, Theorem 1.8]; the exact cokernel comparison is proved here. Corollary 10.3 transports the preceding embedded components to actual section multiplication on $\mathbb{P}^2$.

These results extend, rather than replace, the preceding calculations. The stable generated algebra still carries its representation on the multiplication defect; its exact-action-rank schemes define conductor flags $C \subset Q$ compatible with base change on each stratum (Theorem 6.1). Their rank transitions can themselves be nonreduced. The following determinant-power family provides a complementary, pure-primary case of the theory.

Let Z be the order- h fat point of $\mathbb{P}^e$, with $e \geq 2$ and $h \geq 3$. For every $n \geq 2h-1$, take all series containing $H^0(\mathcal{I}_Z(n))$ and restricting to a generating $(e+1)$ -plane on Z . In symmetric degrees $m \geq h-1$ their failure ideal is

$$\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}, \quad (1.1)$$

where Δ is the integral determinant divisor of the map from the augmentation kernel of the restricted series to $\mathfrak{m}/\mathfrak{m}^2$. This is the full primary ideal, and all its powers remain primary (Corollary 12.3). The proof separates a filtered substitution determinant from a triangular global conductor argument. It gives three-plane results on every $\mathbb{C}[x, y]/(x, y)^h$ and works in arbitrary embedding dimension; no diagonal-ideal theorem is used. The global range depends on the fat-point order, rather than its binomially growing length.

Let $D \subset \mathbb{P}^1$ have length d , put $E_{n,D} = H^0(\mathcal{O}_D(n))$ and $W_D = H^0(\mathcal{O}(n)(-D))$, and let $A \subset E_{n,D}$ be a generating k -plane. Write $U_A = \operatorname{res}_D^{-1}(A)$. For $1 \leq k < d$ our conductor range is $n \geq 2d-k$. In every symmetric degree $m \geq 2$ it gives

$$\operatorname{coker}(\operatorname{Sym}^m U_A \rightarrow V_{mn}) \simeq \operatorname{coker}(\operatorname{Sym}^m \mathcal{A} \rightarrow E_{mn,D}). \quad (1.2)$$

The isomorphism holds on the entire relative generating Grassmannian and after arbitrary base change. The range is sharp uniformly in D, A, m : at $n = 2d-k-1$ an explicit monomial series already violates it in degree two. The exact fibrewise obstruction is the first cohomology of an evaluation kernel. On a curve of genus g , intrinsic cohomological hypotheses give the same transport; the uniform sufficient bound is $\deg L \geq 2d+2g-1$.

The defect-one structural theorem concerns generating hyperplanes in an arbitrary finite locally free algebra, not only pencils or powers of one element. Its failure scheme in all degrees $m \geq 2$ is the space of rank-two quotient algebras with a generating line. The identification is

an isomorphism of functors on arbitrary, including nonreduced, test schemes. It follows from the conductor of a unital hyperplane subalgebra and a cyclic presentation of the multiplication cokernel (Theorem 5.2).

For $D = \sum_i d_i p_i$ this gives a complete primary classification of all generating $(d-1)$ -planes. The components supported on $2p_i$ have transverse algebra

$$G_{d_i} = \mathbb{C}[u, v]/(F_{d_i}, F_{d_i+1}), \quad F_0 = 0, \quad F_1 = 1, \quad F_{j+1} = uF_j + vF_{j-1};$$

those supported on $p_i + p_j$ have transverse algebra $\mathbb{C}[x, y]/(x^{d_i}, y^{d_j})$. Their lengths are $\binom{d_i}{2}$ and $d_i d_j$, but their exact nilpotency indices are $2d_i - 3$ and $d_i + d_j - 1$. The reduced components are disjoint smooth curves on the generating open. The moving scheme is smooth and is a generating open of a projective-line bundle over $\text{Sym}^2 C \times \text{Sym}^{d-2} C$. These results apply to global multiplication on $\mathbb{P}^1$ for $n \geq d+1$. They classify the whole contact-hyperplane family for every multiplicity partition, not a collection of transverse examples.

There is also a complete stable description for generating three-planes on a reduced divisor. On unit frames its ideal is

$$\bigcap_{i < j} (x_i - x_j, y_i - y_j).$$

The equality, and the analogous equality for every power of the ideal, use Haiman's diagonal-ideal theorem explicitly. The step specific to multiplication is an exact total-degree truncation and the conductor comparison; the radical-ideal theorem is a classical input, not a new assertion here.

The pencil family and the distinction between fibres and total spaces. The earlier contact-pencil classification remains part of the article, now in the improved range. We record its complete statement to fix the bases of its primary assertions.

Theorem 1.1 (Contact-pencil failure schemes). *Let $d \geq 4$ and $n \geq 2d - 2$. On the generating open $G_D^\circ \subset \text{Gr}(2, E_{n,D})$, let $\mathcal{J}_m$ be the zeroth Fitting ideal of the global multiplication cokernel. For $2 \leq m < d - 1$, this ideal is zero: the failure scheme is the whole parameter space. For $m \geq d - 1$ it is a Cartier divisor, and, on a fixed divisor $D = \sum_i d_i p_i$, its complete primary decomposition is*

$$\mathcal{J}_m = \bigcap_{i: d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (1.3)$$

Here T_i and E_{ij} are failure to separate $2p_i$ and $p_i + p_j$. These are distinct smooth irreducible divisors; there are no embedded associated points. The exact local equations, all coranks and infinitesimal layers are given in Theorems 14.2 and 14.3. Over a moving multiplicity stratum, primary descent is governed by Lemma 13.1.

With D unrestricted, the pencil failure divisor is integral, and its normalization is the smooth finite incidence marking a scalar length-two subdivisor. The normalization and its singular-support test are given in Theorem 15.2 and Proposition 15.3.

On a unit pencil frame, with $D = (g)$ and ratio v , the determinant is the classical index form $\Delta(g, v) = \det[1, v, \dots, v^{d-1}]$. Its relation to the characteristic polynomial χ_v is

$$\text{disc}(\chi_v) = \text{disc}(g)\Delta(g, v)^2.$$

The equation of failure is $\Delta = 0$, not the discriminant on the left: the latter also vanishes when g has repeated roots, even if v generates the algebra. The quotient and index-form constructions are related to the schemes of generators in [29]. The Grassmannian cohomology presentation of G_r is likewise classical. What must be proved here is the exact identification with the multiplication schemes, including primary fibres and base change.

For the new non-Cartier results the dependence is

$$\begin{aligned} \text{codimension stabilization} &\implies \text{full degree-three Fitting scheme,} \\ \text{hyperplane conductor incidence} &\implies \text{embedded multigenerator local rings,} \\ \text{regularity and the conormal layer} &\implies \text{global embedded failure schemes.} \end{aligned}$$

The first arrow is valid over nonreduced coefficient rings, the second is a scheme isomorphism on the specified open, and the last is a coherent-cokernel isomorphism. These are different kinds of assertions.

The dependence of the retained arguments is

$$\begin{aligned} \text{evaluation kernel} &\implies \text{conductor cokernel,} \\ \text{unital hyperplane conductor} &\implies \text{rank-two quotients,} \\ \text{rank-two quotients} &\implies \text{primary fibres and smooth total scheme,} \\ \text{index form} &\implies \text{pencil primary fibres and normalization,} \\ \text{polarized Cayley-Hamilton + Haiman} &\implies \text{three-plane schemes.} \end{aligned}$$

None of these arguments depends on the statistical interpretation, the quadratic excess classification, or the unrestricted hyperplane embedded structure. Those developments, including all their proofs and hypotheses, are retained in the full complementary manuscript. The present family is not identified with the unrestricted ambient Grassmannian. Section 17 gives the precise classical inputs and the remaining limitation in the nearest-source comparison.

Primary geometry across the conductor boundary. The principal boundary theorem is a classification of an entire Hilbert-function class, rather than a further isolated local example. Every complex local algebra with Hilbert function $(1, 2, 2)$ is isomorphic to $\mathbb{C}[x, y]/((x, y)^3, y^2)$ or $\mathbb{C}[x, y]/((x, y)^3, xy)$. For all its unital three-planes, in every degree at least two, Theorem 8.2 determines the full zeroth-Fitting scheme and every associated stratum, including the extreme-corank boundary. On a smooth frame cover put $\delta = ad - bc$, $P = (a, c)$ and $Q = (b, d)$. In the square case the ideal is $\delta^2 P^2 = (\delta^2) \cap P^4$. In the two-factor case it is $\delta^2(ab, ad + bc, cd)$, with embedded primes $P, Q, P + Q$ and the explicit primary decomposition (8.7). Both schemes have a normal reduced determinant divisor, generic multiplicity two, and nilradical index four.

The factor lines of the quadratic relation index the embedded surfaces. A double factor gives a surface contained entirely in the subalgebra locus; distinct factors give two surfaces meeting that locus and an additional embedded point at the quadratic layer. Theorem 8.3 computes the conductor kernel on these strata, over their actual coefficient rings. This is the boundary information not supplied by the corank-one incidence isomorphism alone. Proposition 8.4 places both types in one finite flat algebra family, with an exact formula for the full Fitting ideal before specialization.

Three levels of statement should be distinguished. For an arbitrary finite algebra, Theorem 7.1 is a global presentation and an incidence theorem. For cube-zero augmentation ideals, Theorem 8.1 factors the full multiplication ideal through the quadratic multiplication tensor. For Hilbert function $(1, 2, 2)$, Theorem 8.2 converts this factorization into a complete primary classification. The connected higher-contact family of Theorem 9.1 supplies a complementary phenomenon: unbounded nilpotent complexity. Corollary 8.5 transports the complete length-five classification to section multiplication on the projective plane.

Finally, Proposition 4.2 proves optimality of the stabilization bound in every codimension and every generating rank at least two. Its elementary field-level interpretation is separated from the coefficient-ring assertion. The arguments in Sections 7.1 and 10.1 give the full functorial and cohomological details needed for arbitrary, possibly nonreduced, base change.

2. SHARP CONDUCTOR REDUCTION AND A CURVE CRITERION

All schemes in this article are over $\mathbb{C}$. Put $V_r = H^0(\mathbb{P}^1, \mathcal{O}(r))$, with $V_r = 0$ for $r < 0$. For a subspace U of sections, U^j denotes its multiplication image, not its abstract symmetric power.

Lemma 2.1 (The evaluation-kernel bound). *Let $U \subset V_n$ be base-point-free, of codimension c , and write $M_U = \ker(U \otimes \mathcal{O} \rightarrow \mathcal{O}(n))$. There are integers $a_i \geq 1$, with $\sum_i a_i = n$, such that $M_U = \bigoplus_{i=1}^{n-c} \mathcal{O}(-a_i)$. If $a = \max_i a_i$, then for $t \geq 0$,*

$$\text{coker}(U \otimes V_t \rightarrow V_{n+t}) \simeq H^1(M_U(t)), \quad UV_t = V_{n+t} \iff t \geq a - 1.$$

In particular $a \leq c + 1$, so $UV_t = V_{n+t}$ for $t \geq c$.

Proof. The splitting theorem for vector bundles on $\mathbb{P}^1$ applies to the locally free evaluation kernel. Its degree is $-n$ and its rank is $n - c$. Moreover $H^0(M_U) = 0$, because $U \rightarrow H^0(\mathcal{O}(n))$ is injective. Thus all its summands have negative degree. Twisting the evaluation sequence by $\mathcal{O}(t)$ and using $H^1(\mathcal{O}(t)) = 0$ gives the asserted cokernel. The formula for $H^1(\mathcal{O}(t - a_i))$ gives the equivalence. Finally the other $n - c - 1$ positive integers have sum at least $n - c - 1$, whence $a \leq n - (n - c - 1) = c + 1$. $\square$

Theorem 2.2 (Sharp uniform conductor range). *Let $D \subset \mathbb{P}^1$ have length $d \geq 2$, let $1 \leq k < d$, and suppose*

$$n \geq 2d - k. \quad (2.1)$$

Let $A \subset H^0(\mathcal{O}_D(n))$ be a generating k -plane, and put $W = H^0(\mathcal{O}(n)(-D))$ and $U = \text{res}_D^{-1}(A)$. For every $m \geq 2$,

$$WU^{m-1} = H^0(\mathcal{O}(mn)(-D)). \quad (2.2)$$

On the entire relative generating Grassmannian over $\text{Hilb}^d(\mathbb{P}^1)$, restriction gives

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn} \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_D(mn)). \quad (2.3)$$

These are isomorphisms of coherent sheaves after arbitrary base change; all their Fitting ideals agree. For a particular geometric fibre, the weaker condition $n - d \geq \max_i a_i(U) - 1$ suffices for the same conclusion. Write $X_{d,k}^\circ$ for this relative generating Grassmannian. The bound (2.1) is sharp as a bound uniform in D, A and simultaneously in all $m \geq 2$. For $k = d$ the cokernel assertion is immediate from $U = V_n$ in the range $n \geq d - 1$; the saturation assertion in that case holds for $n \geq d$.

Proof. Write $D = (g)$, so $W = gV_{n-d}$. The series U is base-point-free: W generates away from D and A generates on D . Its codimension in V_n is $c = d - k$. Lemma 2.1 and $n - d \geq c$ give $WU = gV_{2n-d}$. Further multiplication by U uses input degrees $2n - d, 3n - d, \dots$, all at least c , and proves (2.2). The same argument uses $a - 1$ in place of c on a particular fibre.

All restriction targets and kernels are vector bundles: their first cohomology groups vanish in this range. The map $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U} \rightarrow \pi_* I_D(mn)$ is onto on every geometric fibre, hence is a surjection of vector bundles. It remains onto after every base change. Locally split $0 \rightarrow \mathcal{W} \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$. The kernel of the map on symmetric powers is the image of $\mathcal{W} \otimes \text{Sym}^{m-1} \mathcal{U}$. Quotienting the restriction square by these kernels proves (2.3). Fitting base change now applies [25].

For sharpness take $n = 2d - k - 1$, $D = d\{0\}$, and, in the affine coordinate t , take the degree- n homogenizations of

$$U = \langle 1, t, \dots, t^{k-1}, t^d, t^{d+1}, \dots, t^n \rangle.$$

This is base-point-free and is the inverse image of a generating k -plane on D . Exponents of its quadratic products lie in

$$[0, 2k - 2] \cup [d, n + k - 1] \cup [2d, 2n].$$

None is $2d - 1$. Thus $t^{2d-1} \notin U^2$, although it belongs to $H^0(I_D(2n))$. Its nonzero cokernel class restricts to zero on D , so even the cokernel comparison fails at this boundary in degree two. This does not assert a sharp threshold separately for each $m > 2$. $\square$

The numerical bound can be replaced by a cohomological hypothesis on a curve. Here and below an ideal such as I_D is the ideal on the curve, whereas a Fitting ideal is on the parameter space.

Proposition 2.3 (An exact cohomological test). *Let C be a smooth projective curve, let L be a line bundle, and let $U \subset H^0(L)$ be base-point-free. Put $W = H^0(L(-D))$. If $H^1(L(-D)) = 0$, then*

$$\text{coker}(W \otimes U \rightarrow H^0(L^2(-D))) \simeq H^1(M_U \otimes L(-D)).$$

More generally, for $B_j = U^j \subset H^0(L^j)$ and its evaluation kernel M_{B_j} , the obstruction to $WB_j = H^0(L^{j+1}(-D))$ is exactly $H^1(M_{B_j} \otimes L(-D))$.

Proof. Tensor the evaluation sequence for B_j by $L(-D)$ and take cohomology. The next term is $B_j \otimes H^1(L(-D)) = 0$. $\square$

Theorem 2.4 (Conductor transport on curves). *Let C have genus g , and let D be effective of length $d \geq 2$. Suppose $L(-D)$ is globally generated, $\text{Sym}^2 H^0(L(-D)) \rightarrow H^0(L^2(-2D))$ is surjective, and*

$$H^1(L^j(-2D)) = 0 \quad (j \geq 1). \quad (2.4)$$

For every generating $A \subset H^0(L|_D)$ and every $m \geq 2$, $U = \text{res}_D^{-1}(A)$ satisfies

$$WU^{m-1} = H^0(L^m(-D)), \quad \text{coker}(\text{Sym}^m U \rightarrow H^0(L^m)) \simeq \text{coker}(\text{Sym}^m A \rightarrow H^0(L^m|_D)).$$

In particular these assertions hold if

$$\deg L \geq 2d + 2g - 1. \quad (2.5)$$

For the relative assertion, let $f : C \rightarrow S$ be smooth and projective of relative dimension one, with S locally Noetherian over $\mathbb{C}$, let D be a relative effective Cartier divisor finite flat of degree d , and let L be invertible. Assume the preceding hypotheses on every geometric fibre. For each $j \geq 1$ and $b = 0, 1, 2$, put $E_{j,b} = f_ L^j(-bD)$. These are locally free, $R^1 f_* L^j(-bD) = 0$, and their formation commutes with arbitrary base change. The same base-change property holds for the finite direct images of $L^j|_D$ and $L^j(-D)|_D$. On the relative generating Grassmannian form $\mathcal{U}$ as the inverse image of $\mathcal{A}$ in $E_{1,0}$. The displayed saturation and cokernel assertions then hold as maps of these bundles and sheaves, after every base change.*

Proof. The hypotheses imply $H^1(L^j(-D)) = 0$ by the exact sequence with quotient on D . They make U base-point-free. It contains a pencil without base points, obtained by avoiding the finitely many zeros of one section. The pencil sequence, tensored by $L^j(-2D)$, shows that multiplication onto $H^0(L^{j+1}(-2D))$ is surjective for $j \geq 2$: its obstruction is $H^1(L^{j-1}(-2D))$. Starting with the assumed surjectivity for W^2 , this proves $W^2 U^{m-2} = H^0(L^m(-2D))$.

The map $W \rightarrow H^0(L(-D)|_D)$ is onto by (2.4). Choose $a \in A$ a unit on D and lift it to $u \in U$. Multiplication by u^{m-1} identifies the first normal quotients in these degrees. Consequently Wu^{m-1} fills that quotient in $H^0(L^m(-D))$. Together with the double-vanishing space just proved, this gives the assertion about WU^{m-1} . The locally split restriction square gives the cokernel isomorphism as before.

Under (2.5), the line bundle $L(-D)$ has degree at least $2g + 1$. The classical normal-generation theorem in this degree range gives the required quadratic surjectivity and global generation; see [14, p. 73, Introduction]. Serre duality gives (2.4), since $\deg L(-2D) \geq 2g - 1$. For the relative assertion, the assumed fibrewise H^1 -vanishing for $b = 2$ implies the same vanishing for $b = 1, 0$ by restriction to D . Cohomology and base change for these flat line bundles on a proper flat

relative curve gives the named direct images, with no higher cohomology; finite flat pushforward gives the two restriction bundles. In particular $0 \rightarrow E_{1,1} \rightarrow E_{1,0} \rightarrow f_*(L|_D) \rightarrow 0$ is a locally split exact sequence. The quadratic map $\text{Sym}^2 E_{1,1} \rightarrow E_{2,2}$ and each map $E_{1,1} \otimes \text{Sym}^{m-1} \mathcal{U} \rightarrow E_{m,1}$ are surjective on every geometric fibre by the fixed-curve proof. They are thus surjections of vector bundles and remain so under arbitrary pullback. The same locally split restriction square proves the relative cokernel assertion; no base-change assumption on an unflattened image U^j is needed. $\square$

For example on any elliptic curve, a line bundle of degree at least $2d + 1$ transports all the finite-contact schemes below. The curve statement is not a sharpness assertion in positive genus. Its intrinsic hypotheses, and the exact obstruction in Proposition 2.3, distinguish a failure of regularity from a failure of the finite-contact multiplication.

3. GENERATING OPENS AND FINITE PRESENTATIONS

Here “generating” refers to generation as an ideal or invertible module. It does not assert generation as a unital algebra.

Lemma 3.1 (The relative generating open). *Let B/S be a finite locally free commutative algebra over a $\mathbb{C}$ -scheme, and let $A \subset B$ be a subbundle. The following conditions are equivalent: $B \otimes A \rightarrow B$ is surjective; on every geometric fibre A has nonzero image in every residue field of B ; and the open $A^\times$ of sections of A which are units of B is smooth and surjective over S . These conditions define an open in the Grassmannian. The assertions have the same form for a subbundle of an invertible B -module, using generating sections in place of units.*

Proof. Surjectivity is equivalent to its geometric-fibre version by Nakayama’s lemma. On a finite algebra over an algebraically closed field, an ideal is the unit ideal exactly when its images in all residue fields are nonzero. These evaluations cut out finitely many proper hyperplanes in A . Their complement is nonempty over an infinite field and consists precisely of units. On the total space of the vector bundle A , invertibility is the open condition $\det(b \mapsto ab) \neq 0$. This open is smooth over S , and the preceding fibre calculation proves surjectivity. Conversely a unit section after a faithfully flat cover generates the unit ideal, which descends. The Grassmannian open is the complement of the support of the cokernel of $B \otimes A \rightarrow B$. Local trivializations of an invertible module prove the last assertion. $\square$

Lemma 3.2 (Total-degree truncation). *Let B be a rank- d finite locally free algebra over a $\mathbb{C}$ -algebra R , and let $b_1, \dots, b_q \in B$. The R -subalgebra they generate is spanned as an R -module by the monomials of total degree at most $d - 1$. This assertion commutes with base change, with polynomial reduction coefficients on a free chart.*

Proof. Set $w = \sum_j t_j b_j$ in $B[t_1, \dots, t_q]$ and apply Cayley–Hamilton to multiplication by w . The coefficient of t^α , for $|\alpha| = d$, in w^d is $\binom{d}{\alpha} b^\alpha$. All other terms have B -monomial degree less than d : coefficients of the characteristic polynomial are scalars in $R[t]$. The multinomial coefficient is invertible in R . This expresses each degree- d monomial using lower degrees. Multiplying these identities and inducting reduces every higher monomial. The adjugate identity for Cayley–Hamilton makes the coefficients polynomial in the multiplication table. $\square$

4. A CODIMENSION BOUND OVER THE COEFFICIENT RING

The degree needed to generate a finite algebra can be bounded by the codimension of the initial subspace, rather than by the length of the algebra. A fibrewise dimension argument suggests this bound but does not prove the assertion over a nonreduced parameter scheme. The following argument proves an equality of submodules over the coefficient ring.

Theorem 4.1 (Codimension stabilization). *Let B be a commutative finite locally free algebra over a $\mathbb{Q}$ -scheme, and let $W \subset B$ be a subbundle containing 1, with B/W locally free of rank r . Write W^j for the image of $\text{Sym}^j W \rightarrow B$. Then*

$$W^{r+1} = \mathcal{O}[W], \quad W^j = W^{r+1} \quad (j \geq r+1). \quad (4.1)$$

These equalities remain true after every base change, without a flatness hypothesis on W^j .

For a generating rank- k subbundle A of a rank- d algebra over a $\mathbb{C}$ -scheme, all Fitting ideals of the multiplication cokernel are consequently independent of the degree $j \geq d - k + 1$. The generating opens and normalization by units are those of Lemma 3.1. This is a statement about ideal equalities; the unnormalized cokernels need not have a preferred isomorphism in different degrees.

Proof. The case $r = 0$ is immediate. Work otherwise on a free chart and choose a splitting $B = W \oplus M$, where $\text{rank } M = r$. For $w \in W$ write multiplication by w in blocks

$$L_w = \begin{pmatrix} A_w & D_w \\ \beta_w & C_w \end{pmatrix}.$$

Thus $\beta_w : W \rightarrow M$ and $C_w : M \rightarrow M$. Let $N_0 = \sum_{w \in W} \text{im } \beta_w$, and, for $j \geq 0$, let N_j be the sum of the images of N_0 under all words of length at most j in the operators C_w . A finite local basis of W suffices in these definitions. Since $1 \in W$, every W^m contains W . Projection to M and subtraction of its W component therefore give, as submodules of the chosen splitting,

$$W^{j+2} = W \oplus N_j. \quad (4.2)$$

Indeed the initial case is the definition of N_0 , and multiplication by W changes the residual module N into $N_0 + \sum_w C_w N$.

Commutativity of the full multiplication operators gives the identity

$$[C_w, C_v] = \beta_v D_w - \beta_w D_v, \quad \text{im}[C_w, C_v] \subset N_0. \quad (4.3)$$

Consider a word of length r applied to N_0 . Interchanging two adjacent factors changes it by a word with a commutator in those positions. The commutator sends its entire argument into N_0 ; at most $r - 2$ factors remain to its left. The change thus belongs to N_{r-2} , and in particular to N_{r-1} . For $r = 1$ no interchange is needed. All permutations of a length- r word therefore agree modulo N_{r-1} when applied to N_0 .

Apply Cayley–Hamilton to the r by r matrix $C(t) = \sum_{i=1}^r t_i C_{w_i}$ and extract the coefficient of $t_1 \cdots t_r$. The leading term is the sum of the $r!$ ordered products of the C_{w_i} . Every other term is a scalar linear combination of words of length at most $r - 1$. Applied to N_0 these other terms lie in N_{r-1} . The preceding interchange argument and the invertibility of $r!$ now show that every length- r word sends N_0 into N_{r-1} . Hence $N_r = N_{r-1}$. This equality says that N_{r-1} is invariant under all C_w , so the residual filtration is constant thereafter. Equation (4.2) proves (4.1).

This calculation used no division except by $r!$ and no passage to residue fields. It is valid over every base-changed coefficient ring; in particular, it does not assert that an unsplit image is a vector bundle. Finally choose a unit $a \in A$ smoothly locally and put $W = a^{-1}A$. Multiplication by a^j identifies W^j with A^j , and is an automorphism of B . The Fitting ideals are thus constant in the stated range on a faithfully flat cover and hence on the original parameter scheme. $\square$

Combining this theorem with Lemma 3.2 gives the stable range $j \geq \min\{d - 1, d - k + 1\}$ for $d \geq 2$. The improvement is largest when the starting subspace has small codimension. In particular, a codimension-two multiplication problem is determined in degree three, however large d is. Section 7 uses this ring-level assertion to retain the entire failure scheme, including the closed locus where the defect is two.

4.1. Sharpness and the algebra-length problem. For a unital subspace W over a field, let $\ell(W)$ be the least j such that W^j is the algebra generated by W . This is the usual length of a generating subspace. Algebra-length bounds are studied in the literature of Pappacena and Markova; see [36, 37] and the more recent discussion in [38, Sections 1 and 6]. No novelty is claimed here for the following field-level dimension argument, which works even without commutativity. If $W^j = W^{j+1}$, multiplication by W stabilizes that space and the sequence is constant thereafter. Until then its dimension strictly increases. Starting with dimension k in an algebra of dimension d gives $\ell(W) \leq d - k + 1$. Theorem 4.1, by contrast, proves an equality of submodules over a coefficient ring without assuming the intervening images flat. A dimension count on residue fields does not establish that equality. Its application here is stability of all Fitting ideals after arbitrary base change.

Proposition 4.2 (Sharpness in every codimension). *For every pair of integers $r \geq 1$, $k \geq 2$, there is a connected commutative local algebra B of dimension $r + k$ and a unital k -plane $W \subset B$ such that*

$$W^r \neq W^{r+1} = B.$$

Thus $r + 1$ is the optimal universal bound for codimension r , even when the generating rank k is prescribed.

Proof. Take

$$B = \mathbb{C}[z, \epsilon_1, \dots, \epsilon_{k-2}]/(z^{r+2}, z\epsilon_i, \epsilon_i\epsilon_j : 1 \leq i, j \leq k-2)$$

and $W = \text{span}(1, z, \epsilon_1, \dots, \epsilon_{k-2})$. When $k = 2$ the lists involving ϵ_i are empty. A basis of B is $1, z, \dots, z^{r+1}, \epsilon_1, \dots, \epsilon_{k-2}$. For $1 \leq j \leq r + 1$, a basis of W^j is $1, z, \dots, z^j, \epsilon_1, \dots, \epsilon_{k-2}$. The assertions follow, with z^{r+1} witnessing strictness. $\square$

The result is an optimal bound depending on the initial codimension, not a claim to improve every bound involving the minimal-polynomial degrees or the internal structure of a particular algebra. The relative proof uses commutativity of the full multiplication operators; it does not assert that their compressed blocks commute.

5. GENERATING HYPERPLANES AND LENGTH-TWO QUOTIENTS

We now solve a multigenerator problem without assuming that the finite algebra is monogenic, reduced, or constant. If B is a finite locally free commutative algebra of rank d over a scheme S , let $\text{Hilb}^2(B/S)$ denote the functor of quotient algebras $B_T \rightarrow Q$ that are locally free of rank two over T . It is the relative Hilbert scheme of two points of $\text{Spec}_S B$. A generating line in Q means a line subbundle whose Q -span is Q .

Lemma 5.1 (The conductor of a unital hyperplane). *Unital subalgebras $S_0 \subset B_T$ which are hyperplane subbundles are naturally equivalent to rank-two quotient algebras of B_T . The maps are*

$$S_0 \longmapsto B_T/J, \quad J = \{b \in B_T : bB_T \subset S_0\}, \quad (B_T \rightarrow Q) \longmapsto \text{preimage}(\mathcal{O}_T \cdot 1).$$

The conductor J is an ideal subbundle of rank $d - 2$, and its formation commutes with arbitrary base change.

Proof. Work locally with $T = \text{Spec } R$, trivialize B/S_0 , and write it as the target of $\phi : B \rightarrow R$. Choose e with $\phi(e) = 1$. Since $1 \in S_0$, the bilinear form $\beta(b, c) = \phi(bc)$ has $\beta|_{S_0 \times S_0} = 0$ and $\beta(1, e) = 1$. An element of its radical first lies in S_0 , by pairing with 1; it then lies in the kernel of the split surjection

$$S_0 \longrightarrow R, \quad s \longmapsto \phi(se),$$

which sends 1 to 1. Thus the radical has rank $d - 2$, and B/J has basis the classes of $1, e$. Associativity shows that the radical is an ideal: $\beta(jb, c) = \beta(j, bc) = 0$. It is precisely the displayed conductor. This split-kernel description proves base change and gives $S_0/J = R \cdot 1$.

Conversely, in a rank-two locally free algebra Q , the unit is a unimodular vector, so $Q/R \cdot 1$ is a line bundle. The inverse image of the scalars is a unital hyperplane subalgebra. No nonzero ideal of Q can be contained in $R \cdot 1$: in a basis $1, e$, if $c \cdot 1$ belongs to such an ideal, then ce must be scalar, so $c = 0$. Its conductor is therefore exactly the original kernel. The two constructions are inverse and are independent of all trivializations. $\square$

Theorem 5.2 (The universal hyperplane failure scheme). *Let B/S be finite locally free of rank $d \geq 3$ over a $\mathbb{C}$ -scheme. On $X = \mathrm{Gr}_S(d - 1, B)^\circ$, the generating open, let $\mathcal{A}$ be the tautological hyperplane and put*

$$\mathcal{C}_m = \mathrm{coker}(\mathrm{Sym}^m \mathcal{A} \rightarrow B), \quad Y_m = V(\mathrm{Fitt}_0 \mathcal{C}_m).$$

For every $m \geq 2$ the ideal of Y_m is the same. The common scheme Y is naturally isomorphic to

$$\{(B \twoheadrightarrow Q, \ell) : \mathrm{rank} Q = 2, \ell \subset Q \text{ a generating line}\}. \quad (5.1)$$

The map to X sends (Q, ℓ) to the inverse image of ℓ . If $i : Y \hookrightarrow X$ and $B_Y \twoheadrightarrow Q \supset \ell$ is the universal quotient with its generating line, the canonical formula is

$$\mathcal{C}_m \simeq i_*(Q/\ell^{(m)}), \quad \ell^{(m)} = \mathrm{im}(\ell^{\otimes m} \rightarrow Q). \quad (5.2)$$

Both $\ell^{(m)}$ and its quotient are line bundles on Y . In particular all positive-index Fitting ideals are the unit ideal. The isomorphism of schemes and all these assertions commute with arbitrary base change. The same statement holds for an invertible B -module in place of B , with target its m th tensor power.

Proof. Locally for the smooth topology, choose a unit $a \in \mathcal{A}$. Such choices cover X : on each geometric fibre avoid the finitely many evaluation hyperplanes. Replace $\mathcal{A}$ by $A' = a^{-1}\mathcal{A}$, and the target by multiplication by a^{-m} . Now $1 \in A'$. Split $B = A' \oplus Re$ locally. Since $(A')^m$ contains A' , elementary row and column operations present the cokernel by a single row. For $m = 2$ its entries are exactly the coefficients of the products ss' in B/A' , for $s, s' \in A'$. Call their ideal I . Thus Y_2 represents the condition that A' is a subalgebra, with its full scheme structure, not just its support. Modulo I all higher products lie in A' . Conversely every quadratic product is an m -fold product by inserting 1's. Hence the residual ideals equal I for every $m \geq 2$, and the cokernel is locally R/I .

Lemma 5.1 supplies the unique rank-two quotient on Y . Its kernel J also lies in $\mathcal{A} = aA'$, since J is an ideal and $aJ = J$. The image of $\mathcal{A}$ is a generating line in Q . The construction does not depend on a . Indeed another unit $b \in \mathcal{A}$ gives $b/a \in A'$, and its inverse belongs to A' by Cayley–Hamilton in B (the constant coefficient is a unit). Thus $(b/a)^{-1}A' = A'$ and the conductor is unchanged. This also proves descent from the smooth cover. Conversely the inverse image of a generating line in a rank-two quotient becomes the inverse image of the scalars after division by a unit, so it satisfies precisely the residual equations. These constructions are inverse on every test scheme, proving the scheme isomorphism. On Y , the normalized subalgebra A' has conductor J , and $J \subset a^m A' = (\mathcal{A})^m$ since a is a unit and J an ideal. Passing to Q identifies $(\mathcal{A})^m/J$ with $\ell^{(m)}$. This gives the canonical map in (5.2); the local cyclic presentations identify it as an isomorphism. Multiplication by a local unit spanning ℓ makes $\ell^{(m)}$ a direct-summand line, so the quotient is indeed a line bundle. For an invertible B -module P , put $P_Q = P \otimes_B Q$; the same formula is $i_*(P_Q^{\otimes m}/\ell^{(m)})$, where now $\ell \subset P_Q$. Trivialization of P gives the local proof and this intrinsic expression supplies its gluing. $\square$

5.1. **Every fixed multiplicity partition.** For $r \geq 2$ define polynomials

$$F_0 = 0, \quad F_1 = 1, \quad F_2 = u, \quad F_{j+1} = uF_j + vF_{j-1}, \quad G_r = \mathbb{C}[u, v]/(F_r, F_{r+1}). \quad (5.3)$$

For positive r, s put $H_{r,s} = \mathbb{C}[x, y]/(x^r, y^s)$.

Theorem 5.3 (Complete primary schemes for contact hyperplanes). *Let $D = \sum_i d_i p_i$ have length $d \geq 3$. Consider the entire generating open $\text{Gr}(d-1, H^0(\mathcal{O}_D))^\circ$. For every $m \geq 2$, its multiplication failure scheme is the disjoint union*

$$Y_D \simeq \coprod_{i:d_i \geq 2} (\mathbb{A}^1 \times \text{Spec } G_{d_i}) \amalg \coprod_{i < j} (G_m \times \text{Spec } H_{d_i, d_j}). \quad (5.4)$$

The identifications use local coordinates on D and trivializations of the restricted line bundle. Each displayed summand is a primary component on this fixed-divisor parameter space. There are no embedded associated points. All reduced components are smooth and pairwise disjoint. Their transverse lengths and exact nilradical indices are, respectively,

length-two support	length	nilpotency index
$2p_i$	$\binom{d_i}{2}$	$2d_i - 3$
$p_i + p_j$	$d_i d_j$	$d_i + d_j - 1$.

The convention for a reduced component is index one. Every failure fibre has cokernel dimension one. On binary inverse-image subseries these are exactly the multiplication failure schemes whenever $n \geq d + 1$; on curves they are the same schemes under Theorem 2.4.

Proof. A degree-two quotient of the product algebra $\prod_i \mathbb{C}[z_i]/(z_i^{d_i})$ decomposes according to the ranks of its idempotent factors. These ranks are locally constant even on a nonreduced base. Its possibilities are rank two at a single factor, or rank one at each of two distinct factors. This is a decomposition into open and closed Hilbert subschemes, not merely a decomposition of closed points.

For rank two in $\mathbb{C}[z]/(z^r)$, the elements $1, z$ form a basis in every geometric quotient, hence on every family by Nakayama. Write $z^2 = uz + v$. Then

$$z^j = F_j(u, v)z + vF_{j-1}(u, v).$$

The relation $z^r = 0$ is equivalent to $F_r = vF_{r-1} = 0$, equivalently $F_r = F_{r+1} = 0$. Thus this Hilbert scheme is $\text{Spec } G_r$. A rank-one quotient at a factor of length r is $\text{Spec } \mathbb{C}[x]/(x^r)$, so the two-factor case is $\text{Spec } H_{r,s}$. A generating line in the first rank-two algebra is uniquely spanned by $1 + \tau z$; in the split two-factor algebra it is uniquely spanned by $(1, t)$, with t a unit. Theorem 5.2 now proves (5.4) over arbitrary test schemes.

The algebra G_r is Artin local: if $v \neq 0$, two consecutive zero values in the recurrence force $F_1 = 0$, and if $v = 0$ they force $u = 0$. The two equations form a regular sequence in $\mathbb{C}[u, v]$. With $\deg u = 1$, $\deg v = 2$, its Hilbert series is

$$\frac{(1 - t^{r-1})(1 - t^r)}{(1 - t)(1 - t^2)}. \quad (5.5)$$

It has length $\binom{r}{2}$ and weighted socle degree $2r - 4$. To see the exact ordinary nilpotency, identify u and $-v$ with the elementary symmetric functions in two variables. Then $F_j = h_{j-1}$ and

$$G_r \simeq H^{2*}(\text{Gr}(2, r), \mathbb{C}), \quad (5.6)$$

where the cohomological degree is divided by two. This is the classical presentation by h_{r-1}, h_r , not a new presentation of Grassmannian cohomology; the rectangular Schur basis is recalled in [31, Theorem 2.7 and the following discussion]. Successive multiplication by $u = s_{(1)}$ adds one

box. The only partition of size $2r - 4$ in the 2 by $r - 2$ rectangle is $(r - 2, r - 2)$; the ballot-path count of its standard tableaux is $\frac{1}{r-1} \binom{2r-4}{r-2}$. Consequently

$$u^{2r-4} = \frac{1}{r-1} \binom{2r-4}{r-2} (-v)^{r-2} \neq 0.$$

Every monomial of ordinary degree $2r - 3$ has weighted degree at least $2r - 3$ and vanishes. This proves the index $2r - 3$, including $r = 2$. In $H_{r,s}$ the top monomial is $x^{r-1}y^{s-1}$, giving length rs and index $r + s - 1$.

Each summand is a complete intersection with reduced support a smooth integral curve, so it is primary and has no embedded points. Equivalently, over a local Artin coefficient algebra, a polynomial with nonzero reduction is a nonzerodivisor, as one checks on the successive powers of the maximal ideal. The same holds after inverting t . This proves the primary assertions also globally on the indicated affine curves. The cokernel assertion follows from Theorem 5.2. The two transports follow from Theorems 2.2 and 2.4. $\square$

Theorem 5.4 (The smooth moving scheme). *Let $d \geq 3$. On the relative generating Grassmannian of $(d - 1)$ -planes over $\text{Hilb}^d(C)$, for a smooth connected projective curve C , the finite-contact failure scheme is a generating open of a projective-line bundle over*

$$\text{Sym}^2 C \times \text{Sym}^{d-2} C, \quad (Z, Q) \mapsto D = Z + Q.$$

It is smooth, irreducible, and of dimension $d + 1$. Its projection to $\text{Sym}^d C$ is flat of relative dimension one. The finite map of divisor spaces has degree $\binom{d}{2}$; the primary lengths in Theorem 5.3 sum to this degree on every fixed fibre. Under either conductor theorem, these are statements about the original global multiplication scheme.

Proof. A nested pair $Z \leq D$ of effective divisors on a smooth curve is uniquely $D = Z + Q$. This holds in families: a finite flat subscheme of a smooth relative curve is a relative Cartier divisor, and inclusions of their invertible ideals give the residual divisor. The incidence in Theorem 5.2 is therefore exactly the following canonical open projective bundle. On $T = \text{Sym}^2 C \times \text{Sym}^{d-2} C$ let $\mathcal{Z} \subset C \times T$ be the universal degree-two divisor and set $F = (\pi_{\mathcal{Z}})_* \mathcal{O}_{\mathcal{Z}}$. Use the projectivization $\mathbb{P}_T^{\text{lines}}(F)$ parametrizing line subbundles of F , and retain the open where the tautological line ℓ satisfies $F\ell = F$. This is the claimed bundle, not an unspecified local form. For a restricted line bundle replace F by $(\pi_{\mathcal{Z}})_*(L|_{\mathcal{Z}})$ and use generation as an F -module. The open is nonempty on every fibre of its smooth irreducible base, and hence is smooth and irreducible of dimension $d + 1$.

The map of divisor spaces is finite (proper with finite geometric fibres). Both spaces are smooth of dimension d ; the finite source is Cohen–Macaulay and the target regular, so the map is flat. Over a reduced divisor it has $\binom{d}{2}$ points. This proves the degree and flatness assertions; alternatively, the fibre lengths computed above give $\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2}$. The open projective bundle is smooth of relative dimension one, so its composite projection is flat of that relative dimension. $\square$

5.2. A direct residual presentation. For comparison with the multiplication matrix, normalize a unit and write, on the unital hyperplane chart in $\mathbb{C}[z]/(z^r)$,

$$A' = \langle 1, z^i - c_i z : 2 \leq i \leq r - 1 \rangle.$$

Put $c_1 = 1$ and $c_j = 0$ for $j \geq r$. After eliminating the pivot columns, the remaining entries of the quadratic matrix are

$$c_{i+j} - c_i c_{j+1} - c_j c_{i+1} + c_2 c_i c_j \quad (2 \leq i \leq j \leq r - 1). \quad (5.7)$$

For $r \geq 4$, the relations with $i = 2$ give $c_{j+2} = uc_{j+1} + vc_j$, with $u = c_2$, $v = c_3 - c_2^2$. They eliminate $c_4, \dots, c_{r-1}$ and leave F_r, F_{r+1} . Every other relation follows from the recurrence identity

$$F_{i+j} = F_i F_{j+1} + F_j F_{i+1} - u F_i F_j,$$

proved by induction in j . For $r = 3$ the single residual entry is c_2^3 , agreeing with $G_3 \simeq \mathbb{C}[u]/(u^3)$. For $r = 4$ one obtains

$$G_4 = \mathbb{C}[u, v]/(u^3 + 2uv, u^2v + v^2), \quad \text{length } G_4 = 6, \quad \mathfrak{m}^5 = 0 \neq \mathfrak{m}^4.$$

Thus the multigenerator transverse scheme is not a powered smooth divisor, even though its length equals the corresponding pencil weight. The smooth moving scheme has these nonreduced fibres; nonreducedness of a fibre is not an embedded component of the total scheme. Nor does this determine the embedded primary ideal along the evaluation curve in the unrestricted, non-generating hyperplane problem retained in the complementary material.

5.3. A noncurvilinear algebra and an embedded component. The general finite-algebra theorem is not restricted to the primary patterns arising from divisors on a smooth curve.

Proposition 5.5 (An exact embedded chart). *Let $B = \mathbb{C}[x, y]/(x^2, y^2)$. On the unit-normalized hyperplane chart*

$$A' = \langle 1, y - ax, xy - bx \rangle \subset B \otimes \mathbb{C}[a, b],$$

the multiplication failure ideal, for every $m \geq 2$, is

$$I = (ab, b^2) = (b) \cap (a, b)^2.$$

Its associated primes are (b) and (a, b) . Its nilradical is a one-dimensional square-zero module supported at the embedded point. These equations give the pullback of the full failure scheme on the generating Grassmannian, not only its reduced support.

Proof. Use the quotient functional $\phi(1) = 0$, $\phi(x) = 1$, $\phi(y) = a$, $\phi(xy) = b$. Products involving 1 are pivots. The other quadratic products have residual values $-2ab, -b^2, 0$. This gives I ; Theorem 5.2 gives every higher degree. The displayed primary decomposition follows by intersecting the monomial ideals. The nilradical is generated by b , whose annihilator modulo I is (a, b) . The normalization by a unit is the same smooth-cover calculation used in that theorem. In particular adding the freely chosen unit does not remove the embedded associated prime. $\square$

Thus the quotient description distinguishes smooth moving contact schemes from finite-algebra families with embedded structure. The absence of embedded points in Theorem 5.3 is a consequence of the curvilinear contact algebra and its explicit Hilbert schemes, not a general assertion about finite algebras.

6. STABLE IMAGES AND HIGHER-DEFECT CONDUCTOR STRATA

The rank-two quotient in Theorem 5.2 is specific to a hyperplane. In larger codimension the appropriate object is a quotient algebra together with a subalgebra acting faithfully on the quotient module. The rank of this action is an additional stratification invariant. We keep its scheme structure throughout.

For a map u of vector bundles write $I_j(u)$ for its ideal of j -minors. Its *exact-rank s locus* means $V(I_{s+1}(u)) \setminus V(I_s(u))$, with this locally closed scheme structure. On it the image and cokernel are vector bundles and commute with arbitrary base change: after inverting a pivot, elementary operations reduce the matrix to $\text{diag}(I_s, 0)$. This is stronger than requiring constant rank on geometric points.

Theorem 6.1 (Stable image and conductor stratification). *Let B be a commutative finite locally free algebra of rank $d \geq 2$ over a locally Noetherian $\mathbb{C}$ -scheme S , and let $\mathcal{A}$ be the universal bundle on $X = \mathrm{Gr}_S(k, B)^\circ$, with $1 \leq k \leq d$. Put $\mu_m : \mathrm{Sym}^m \mathcal{A} \rightarrow B_X$. All Fitting ideals of $\mathrm{coker} \mu_m$ are independent of $m \geq d - 1$; Theorem 4.1 also gives stability for $m \geq d - k + 1$. We use $d - 1$ below to retain a uniform notation for the stable-image strata. On the exact-rank s locus X_s of μ_{d-1} there is a canonical unital subalgebra bundle $E \subset B_{X_s}$ of rank s . There is an invertible E -module $L \subset B_{X_s}$, also a subbundle over X_s , such that*

$$E = \mathcal{O}[a^{-1}\mathcal{A}], \quad L = E\mathcal{A} = aE, \quad \mathrm{im} \mu_m = L^m \quad (m \geq d - 1) \quad (6.1)$$

on any local unit cover $a \in \mathcal{A} \cap B^\times$.

Suppose $r = d - s > 0$ and put $M = B/E$. The action

$$\theta : E \longrightarrow \mathcal{E}nd_{\mathcal{O}}(M)$$

defines exact-rank strata $X_{s,t}$, with $1 \leq t \leq r^2$. On such a stratum the conductor and its quotient flag are

$$J = \ker \theta = \{b \in B : bB \subset E\}, \quad C = E/J \subset Q = B/J, \quad \mathrm{rank} C = t, \quad \mathrm{rank} Q = r + t. \quad (6.2)$$

Here J is an ideal subbundle of B , and $C \rightarrow \mathcal{E}nd(Q/C)$ is a subbundle injection. Each of these constructions commutes with arbitrary base change within the specified stratum.

More generally, the functor of unital codimension- r subalgebra bundles with action of exact rank t is equivalent to the functor of flags $B \twoheadrightarrow Q \supset C$ of these ranks, where the first map is a surjective morphism of unital commutative algebras, Q is locally free of rank $r + t$, $C \subset Q$ is a unital subalgebra subbundle of rank t , Q/C is locally free of rank r , and the specified multiplication action $C \rightarrow \mathcal{E}nd(Q/C)$ is a locally split subbundle injection. The inverse sends the flag to the preimage of C .

Proof. Choose a unit a smoothly locally, as in Lemma 3.1, and set $W = a^{-1}\mathcal{A}$. Then $1 \in W$. Lemma 3.2 says that $W^{d-1} = \mathcal{O}[W]$ and that $W^m = W^{d-1}$ for $m \geq d - 1$, over the coefficient ring itself. Multiplication by a^m identifies the original image with $a^m\mathcal{O}[W]$. The resulting cokernels are locally isomorphic in these degrees, proving equality of every Fitting ideal before any rank restriction.

On X_s , the pivot description makes $E = \mathcal{O}[W]$ a subbundle with locally free quotient. If b is a second unit in $\mathcal{A}$, then $b/a \in E$ is a unit of B . Cayley–Hamilton for multiplication by b/a , whose constant coefficient is a unit, puts its inverse in E . Thus $\mathcal{O}[b^{-1}\mathcal{A}] \subset E$; reversing a, b gives equality. This proves descent of E . The image $E\mathcal{A}$ is locally aE and therefore descends to the stated invertible E -module and subbundle L . It also proves (6.1) universally.

Because E is a subalgebra, it acts on $M = B/E$. Its action kernel consists of the $e \in E$ with $eB \subset E$. An element b of the displayed conductor already lies in E , by multiplication by 1; hence the two kernels agree. If $jB \subset E$, then for $b, c \in B$ one has $(jb)c = j(bc) \in E$, so J is an ideal of B . On an exact-rank t stratum, pivot reduction for θ makes J locally free of rank $s - t$ and C a subbundle of $\mathcal{E}nd(M)$. The exact sequence $0 \rightarrow C \rightarrow B/J \rightarrow M \rightarrow 0$ proves that Q is locally free of rank $r + t$. All the kernels and images just used are split as $\mathcal{O}$ -modules locally; this proves their base-change assertions. The unit acts as the identity on M , so $t \geq 1$; the other bound comes from $\mathrm{rank} \mathcal{E}nd(M) = r^2$.

Conversely, given a flag as in the statement, its inverse image E is a subalgebra bundle and $B/E \simeq Q/C$. The kernel of its action is precisely the kernel of $B \rightarrow Q$: a larger kernel would give an element of C annihilating Q/C , contrary to the subbundle injection. Thus the conductor construction recovers the flag, including over nonreduced test schemes. These mutually inverse constructions respect all pullbacks within the exact-rank functors. $\square$

The word “stratum” cannot be removed from the base-change assertion for J . The multiplication cokernel is a universally defined coherent sheaf on X ; its conductor need not be a bundle on X . Example 6.5 below exhibits the distinction.

Proposition 6.2 (The extreme-corank failure scheme). *On the Grassmannian of rank- $(d - r)$ subbundles $W \subset B$ containing 1, with $1 \leq r < d$, the unital subalgebra scheme is*

$$V(\text{Fitt}_{r-1} \text{coker}(\text{Sym}^m W \rightarrow B)), \quad m \geq 2.$$

The defining ideal is independent of m , including its nonreduced structure. Its conductor exact-rank t subschemes are the quotient-flag functors of Theorem 6.1.

Proof. Since $1 \in W$, the image W^m contains W . Split $B = W \oplus M$ locally and remove the identity pivot columns. The residual presentation has r rows, so its $(r - 1)$ st Fitting ideal is the ideal of all entries. In degree two these entries are precisely the coefficients of $W \cdot W$ in B/W . They vanish if and only if W is a subalgebra, on every test scheme. Modulo their ideal, every higher product remains in W . Conversely each quadratic product is an m -fold product by insertion of 1’s. The entry ideals are equal. The last assertion is the functorial equivalence just proved. $\square$

For $r = 1$ this is the usual zeroth-Fitting failure scheme. For $r > 1$ it is the *extreme-corank* locus, not the whole locus where multiplication is nonsurjective. This distinction prevents a false extension of the hyperplane theorem by changing only the quotient rank.

6.1. The two quotient mechanisms in codimension two.

Lemma 6.3 (Relative cyclic vectors). *Let M be a rank-two vector bundle, and let $C \subset \mathcal{E}nd(M)$ be a unital rank-two algebra subbundle over a $\mathbb{C}$ -scheme. Then M is an invertible C -module locally on the base. More explicitly, write $C = \langle 1, T \rangle$ and $T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ after trivializing. The opens*

$$D(c), \quad D(b), \quad D(c + d - a - b)$$

cover the base and have cyclic vectors $(1, 0)$, $(0, 1)$, and $(1, 1)$, respectively. These opens and the resulting module isomorphisms commute with base change.

Proof. On every geometric fibre, the subbundle injection makes T nonscalar. If all three displayed functions vanish, then $b = c = d - a = 0$, contradicting that assertion. Their principal opens therefore cover. The determinant of the map $C \rightarrow M$, $q \mapsto qv$, is respectively c , $-b$, and $c + d - a - b$ for the three chosen vectors. On the relevant open it is a unit, so the map is a C -linear isomorphism. The same invertible matrices prove the assertion after any pullback. $\square$

Theorem 6.4 (Codimension-two quotient flags). *Let $E \subset B$ be a unital subalgebra bundle with $M = B/E$ of rank two over a locally Noetherian $\mathbb{C}$ -scheme. For its action θ one has $I_3(\theta) = 0$ scheme-theoretically. The exact-rank-one closed locus and the exact-rank-two open locus have the following functorial descriptions.*

On the first, E is the inverse image of the scalar subalgebra in a locally free rank-three quotient algebra of B . On the second, E is the inverse image of C in a quotient $B \twoheadrightarrow Q$, where C is a rank-two algebra over the base and Q is a locally free rank-two algebra over C . In particular Q has rank four over the base. In both cases the kernel of $B \rightarrow Q$ is the conductor, and these descriptions commute with arbitrary base change on their respective strata.

Proof. Locally trivialize M . Since 2 is invertible, write every matrix as a scalar plus a trace-zero matrix. The trace-zero parts of the commuting matrices $\theta(E)$ still commute. The bracket map

$$\bigwedge^2 \mathfrak{sl}_2 \longrightarrow \mathfrak{sl}_2$$

is an isomorphism over $\mathbb{C}$: for the usual basis H, U, V its values are $2U, -2V, H$. Thus the wedge of any two trace-zero columns is zero over the whole coefficient ring, not just over its residue fields. Adjoining the scalar identity column proves $I_3(\theta) = 0$. The identity also gives $I_1(\theta) = \mathcal{O}$.

On the exact-rank-one locus the image is the scalar algebra. Theorem 6.1 gives $C = \mathcal{O}$ and $\text{rank } Q = 3$. Conversely the scalar line in any rank-three algebra has faithful action on its rank-two quotient module: the action is ordinary scalar multiplication. Its inverse image has exactly this conductor.

On the exact-rank-two locus, $C \subset \mathcal{E}nd(M)$ is a rank-two subalgebra bundle. Locally write $C = \langle 1, T \rangle$. Lemma 6.3 gives an explicit cyclic-vector open cover and proves that M is locally free of rank one as a C -module. The exact sequence of C -modules $0 \rightarrow C \rightarrow Q \rightarrow M \rightarrow 0$ consequently splits locally, so Q is locally free of rank two over C .

Conversely, if Q is a locally free rank-two C -algebra, its unit is unimodular as a C -section. Hence Q/C is locally free of rank one over C . Its regular C -action is faithful and is a subbundle injection into its endomorphism bundle over the original base (check on geometric fibres and use a pivot). The quotient-flag equivalence therefore gives exactly the second stratum. All arguments are functorial and use split modules on the indicated strata. $\square$

Remark 6.5. Consider $R = \mathbb{C}[s]$, $B = R[z]/(z^5)$, and the direct summand

$$E = \langle 1, sz^2 + z^3, z^4 \rangle_R.$$

It is a subalgebra since $(sz^2 + z^3)^2 = s^2 z^4$. Modulo E take the basis $\bar{z}, \bar{z}^2$; then $\bar{z}^3 = -s\bar{z}^2$. The action of $sz^2 + z^3$ sends $\bar{z}$ to $-s^2 \bar{z}^2$ and $\bar{z}^2$ to zero, while z^4 acts as zero. Thus the exact-rank-one subscheme is $s^2 = 0$, not merely $s = 0$. Over R the conductor is Rz^4 ; at the reduced fibre $s = 0$ it is $\langle z^3, z^4 \rangle$. Quotient rank drops from four to three. This simultaneously records the nonreduced rank stratum and the failure of conductor formation to commute with unrestricted specialization. The fibre subalgebras belong to the classical codimension-two polynomial-algebra classification of [34, Theorem 39]; the calculation here retains the relative action and its determinantal thickening.

7. THE FULL CODIMENSION-TWO FAILURE SCHEME

Fix a finite locally free commutative algebra B/S of rank $d \geq 4$ over a locally Noetherian $\mathbb{C}$ -scheme. Let

$$X = \text{Gr}_S(d-3, B/\mathcal{O} \cdot 1),$$

viewed as the Grassmannian of rank- $(d-2)$ subbundles $W \subset B$ containing 1. This is a unital parameter space, not an open subset of the Grassmannian of all generating planes. We explain the passage to the latter at the end of the section.

Write $\mu_j : \text{Sym}^j W \rightarrow B_X$, and let D denote the stable zeroth-Fitting failure scheme. Let $T \subset X$ denote the unital subalgebra scheme. In general T is a proper closed subscheme of D ; its ideal is the first, rather than the zeroth, Fitting ideal.

Theorem 7.1 (Equations and the corank-one incidence). *The stable scheme D is defined already by μ_3 . On a splitting $B_X = W \oplus M$, with $\text{rank } M = 2$, use the block notation of (4.3). Let β be the matrix with columns $\beta_{w_i}(w_j)$, for a local basis (w_i) of W , and set*

$$\mathcal{K} = [\beta \mid C_{w_1}\beta \mid \cdots \mid C_{w_{d-2}}\beta]. \quad (7.1)$$

For every $m \geq 3$ there are equalities

$$\begin{aligned} \text{coker } \mu_m &\simeq \text{coker } \mathcal{K}, \\ \mathcal{I}_D &= I_2(\mathcal{K}), \\ \mathcal{I}_T &= I_1(\beta) = I_1(\mathcal{K}) = \text{Fitt}_1 \text{coker } \mu_m. \end{aligned} \quad (7.2)$$

The cokernel isomorphism uses the chosen splitting; the ideals glue canonically. These are equations for the full schemes, not just for their radicals or their exact-rank strata.

Let $H = \text{Hilb}_S^2(B)$ be the scheme of locally free rank-two quotient algebras. Over H , let $E_H \subset B_H$ be the preimage of the scalar subalgebra in the universal quotient. Put

$$Z = \text{Gr}_H(d-3, E_H/\mathcal{O} \cdot 1), \quad Z^\circ = \{W : W^2 \longrightarrow E_H \text{ is onto}\}.$$

Then the forgetful morphism induces an isomorphism

$$Z^\circ \xrightarrow{\sim} D \setminus |T|. \quad (7.3)$$

It holds on arbitrary test schemes, including nonreduced ones. On this open the stable generated algebra is the hyperplane E_H , and the normalized multiplication cokernel is its quotient line B/E_H . The proper morphism $Z \rightarrow X$ factors through D and is surjective on geometric points of D .

Proof. Theorem 4.1 gives $W^3 = \mathcal{O}[W]$. By (4.2), its residual image in M is $N_1 = \text{im } \mathcal{K}$. The identity columns supplied by $W \subset W^3$ may be removed from a presentation of B/W^3 . This proves the first two assertions in (7.2). Every entry of $C_{w_i}\beta$ is a linear combination of entries of β , so their entry ideals are equal. Vanishing of these entries is exactly $W^2 \subset W$, which is the subalgebra functor. This proves the last line.

Work now where a column v of β is unimodular in M . Its lift is a quadratic product modulo W , and $E = W + \mathcal{O}v$ is a hyperplane subbundle of B . Pivot on the unit coordinate of v . The two by two minors of $\mathcal{K}$ then say precisely that all its columns belong to $\mathcal{O}v$. In the quotient ring by those minors, $W^2 = E$ and $WE \subset E$. Moreover $v \in W^2$, so multiplication by v preserves E : write v as a sum of products of elements of W and use $WE \subset E$ twice. Thus E is a unital subalgebra. Equivalently its equality with the stable image follows directly from Theorem 4.1.

The conductor equivalence of Lemma 5.1 associates to E a unique rank-two quotient of B . It recovers a point of Z° . Conversely, a point of Z° has stable image E_H , since $W^2 = E_H$ and E_H is an algebra. Its residual multiplication rank is exactly one, so it factors through $D \setminus |T|$. The two constructions are inverse on every test scheme and are defined by split pivots and the hyperplane conductor. They therefore glue to (7.3), including its nilpotent structure. The same description gives the cokernel line.

The relative Hilbert scheme is projective: it is the closed ideal-quotient subscheme of the Grassmannian of rank-two module quotients of B . Hence $Z \rightarrow X$ is proper. Every multiplication image on Z lies in the hyperplane algebra E_H , so its maximal minors vanish and the map factors through D . Surjectivity away from T is already proved. Surjectivity on geometric points of T follows from the two fibre descriptions in Proposition 7.2 below. $\square$

The theorem gives both an equation for D along T and a complete functorial normal form off T . In particular, it is not obtained by replacing the ideal $I_2(\mathcal{K})$ with the entry ideal $I_1(\beta)$. Nor does it assert that the corank-one open is dense in every component: for a particular algebra a component may be contained in T .

Proposition 7.2 (Fibres over the conductor strata). *Restrict T to an exact-action-rank stratum and write $J = \text{cond}_B(W)$, $C = W/J$, and $Q = B/J$. The fibre product $Z \times_X T$ has the following descriptions, with arbitrary base change within the indicated stratum.*

If the action rank is one, $C = \mathcal{O}$ and Q has rank three. The fibre product is $\text{Hilb}^2(Q)$. More explicitly, on $\mathbb{P}(Q/\mathcal{O})$, in the convention of lines, it is the zero scheme of the canonical binary cubic

$$\ell^{\otimes 3} \longrightarrow \bigwedge^2(Q/\mathcal{O}), \quad v^{\otimes 3} \longmapsto \bar{v} \wedge \overline{v^2}. \quad (7.4)$$

The cubic may vanish identically; no relative degree-three flatness is asserted without a nonvanishing hypothesis.

If the action rank is two, C has rank two and Q is a rank-two C -algebra. The fibre product is canonically $\mathrm{Spec} C$, a finite locally free double cover of the stratum, possibly nonreduced.

Proof. A point of the fibre product is a hyperplane algebra $E \subset B$ containing W . Since J is an ideal of B contained in W , it is contained in the conductor of E . Thus the problem is exactly to choose a unital hyperplane subalgebra of Q containing C . Conversely its inverse image supplies E .

In the first case this is the unital-hyperplane functor of a rank-three algebra, hence $\mathrm{Hilb}^2(Q)$ by Lemma 5.1. A line $\ell \subset Q/\mathcal{O}$ has an algebra as its preimage if and only if its square remains in that preimage. This is exactly (7.4). Changing a lift v by a scalar changes v^2 by a scalar and a multiple of v , so the displayed wedge is independent of the lift. This also proves its scheme-theoretic meaning over a nonreduced base.

In the second case put $M = Q/C$. By Lemma 6.3, M is locally an invertible C -module. A hyperplane extension gives a C -stable subbundle $\ell \subset M$ of rank one over the original base. The induced action on the quotient line M/ℓ is a character $C \rightarrow \mathcal{O}$. Conversely such a character, with kernel I , gives $\ell = IM$; it is a subbundle and the quotient is a line. These constructions are inverse, as can be checked after trivializing M over C . They are the functor of $\mathrm{Spec} C$. There is no additional multiplication equation. Indeed locally write $Q = C \oplus Cz$, with $z^2 = \alpha + \gamma z$, $\alpha, \gamma \in C$. The extension is $C \oplus Iz$, which is an algebra because I is an ideal of C . This proves the assertion over every test scheme.

For completeness both fibres are nonempty over an algebraically closed field. A rank-three commutative algebra has a quotient of length two, by an ideal composition series; a rank-two algebra has a character. This proves the geometric surjectivity used above. $\square$

For example, the rank-three algebra $\mathbb{C}^3$ gives three points in (7.4), whereas $\mathbb{C}[u, v]/(u, v)^2$ gives the whole projective line. Mere invariance under the scalar action would incorrectly give a projective line in both cases. The square condition in (7.4) is essential. In contrast, on the rank-two action stratum it follows from the quadratic-algebra tower. The conductor stratification thus controls the exceptional fibres of one proper incidence over the full failure scheme. We do not identify this morphism with a normalization or a blow-up.

Remark 7.3 (Unit covers and unrestricted generating planes). Let $G = \mathrm{Gr}_S(d-2, B)^\circ$ be the full generating Grassmannian, and let $\tilde{G}$ parametrize pairs (A, a) with $a \in A \cap B^\times$. The morphism $\tilde{G} \rightarrow G$ is smooth and surjective, and

$$\tilde{G} \simeq B^\times \times_S X, \quad (A, a) \longmapsto (a, a^{-1}A).$$

Multiplication by a^{-m} identifies the pulled-back cokernel with the unital one. All the ideal identities above therefore give the full failure scheme on this smooth cover of G and descend. Over $\mathbb{C}$, local primary structures discussed below may equally be read on this cover; the assertion is not that a special linear slice of G determines its ambient scheme structure.

7.1. The quadratic incidence fibre as a functor. In Theorem 7.1, the notation $D \setminus |T|$ means the open complement of the support of the possibly nonreduced closed scheme T . Nothing is subtracted from its structure sheaf. We supply a full functorial description of the rank-two action fibre in Proposition 7.2.

Let $C \subset Q$ be the conductor flag on that exact-action-rank stratum. Then C is a rank-two algebra, Q is a rank-two locally free C -algebra, and $M = Q/C$ is an invertible C -module. The incidence fibre parametrizes rank-three subalgebras $E' \subset Q$ containing C , with Q/E' a line bundle over the base.

A morphism $U \rightarrow \mathrm{Spec} C$ is an algebra map $\chi : C_U \rightarrow \mathcal{O}_U$. Write $I = \ker \chi$; it is a line subbundle of C_U , since the unit splits χ . Tensoring $0 \rightarrow I \rightarrow C_U \rightarrow \mathcal{O}_U \rightarrow 0$ with the invertible

C_U -module M_U gives

$$0 \longrightarrow IM_U \longrightarrow M_U \longrightarrow M_U \otimes_{C_U, \chi} \mathcal{O}_U \longrightarrow 0.$$

The last term is a line bundle over U . The middle term has rank two over U , so IM_U is a line subbundle and this sequence is locally split over $\mathcal{O}_U$. Let E'_χ be its inverse image in Q_U . It is a rank-three subbundle containing C_U . To check multiplicative closure, one may work faithfully flat locally where M_U has a C_U -generator lifted to $e \in Q_U$. Then $Q_U = C_U \oplus C_U e$, $e^2 = \alpha + \beta e$, and $E'_\chi = C_U + Ie$. Products stay in this submodule because I is an ideal and $I^2 \subset I$. The closure condition descends, and the definition of E'_χ did not use the generator.

Conversely let E' be such a subalgebra, and put $N = E'/C_U$. The exact sequence

$$0 \longrightarrow N \longrightarrow M_U \longrightarrow Q_U/E' \longrightarrow 0$$

exhibits N as a line subbundle. Since $C_U \subset E'$, the submodule N is C_U -stable. Its quotient line $L' = Q_U/E'$ therefore defines an algebra map

$$\chi : C_U \longrightarrow \text{End}_{\mathcal{O}_U}(L') = \mathcal{O}_U.$$

The equality on the right is canonical for an invertible module. Let $I = \ker \chi$. The quotient map factors through $M_U \otimes_{C_U, \chi} \mathcal{O}_U \rightarrow L'$. This is a surjective map of line bundles, hence an isomorphism. Thus $N = IM_U$ and $E' = E'_\chi$. Conversely the action on the quotient constructed from χ recovers χ itself. All the quotient sequences are locally split as sequences of base modules, and tensoring an invertible C -module is exact. The two constructions therefore commute with every pullback and are inverse on every test scheme, not just on geometric points. This proves the asserted fibre isomorphism with $\text{Spec } C$ by the functor of points.

8. PRIMARY GEOMETRY AT A CONDUCTOR BOUNDARY

The distinction between a presentation of the full failure scheme and its primary geometry is essential along the subalgebra locus. We now resolve that geometry for every complex local algebra with Hilbert function $(1, 2, 2)$. The proof begins with a factorization valid for all algebras whose augmentation ideal has cube zero. In the classified case the quadratic relation determines the embedded associated strata, including a stratum supported entirely at the extreme-corank boundary. No restriction to the quadratic-generation open is made.

8.1. The quadratic-layer factorization. Let B be a finite local $\mathbb{C}$ -algebra, with maximal ideal $\mathfrak{m}$ satisfying $\mathfrak{m}^3 = 0$. Put $V = \mathfrak{m}/\mathfrak{m}^2$, $S_2 = \mathfrak{m}^2$, and write $e = \dim V$, $q = \dim S_2$. Multiplication induces a surjection

$$\gamma : \text{Sym}^2 V \longrightarrow S_2.$$

Choose a splitting $\mathfrak{m} = V \oplus S_2$. The unital parameter space of $(e+1)$ -planes is $X = \text{Gr}(e, \mathfrak{m})$: its tautological plane is $W = \mathcal{O} \cdot 1 \oplus K$. On its frame bundle write

$$\begin{pmatrix} M \\ H \end{pmatrix} : \mathcal{O}^e \longrightarrow V \oplus S_2, \quad M \in \text{Mat}_{e \times e}, \quad H \in \text{Mat}_{q \times e}. \quad (8.1)$$

The stacked matrix has rank e . Its entries, not just the entries of M , are the frame coordinates.

Theorem 8.1 (Quadratic-layer factorization). *For every degree $m \geq 2$, the full zeroth-Fitting ideal of $\text{Sym}^m W \rightarrow B$ pulls back to*

$$\mathcal{I}_D = (\det M) I_q(\gamma \text{Sym}^2 M). \quad (8.2)$$

The identity holds on the whole frame bundle and after arbitrary base change, including substitution of coefficients in a nonreduced ring. In particular it also gives equations at $M = 0$ whenever that stratum meets the frame bundle. The variables H are unrestricted except for the frame condition; the formula is not a transverse slice.

Proof. Products of three elements of K vanish. Because $1 \in W$, the multiplication image in every degree $m \geq 2$ is $\mathcal{O} \cdot 1 + K + K^2$. After deleting zero and repeated columns, a presentation of its cokernel is given by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & M & 0 \\ 0 & H & \gamma \operatorname{Sym}^2 M \end{pmatrix}.$$

Every nonzero maximal minor must use the first column and all e columns of the middle block: no other column has an entry in the V rows. Expansion in the first $e + 1$ rows leaves exactly $\det M$ times a q by q minor of the last block. These are all the maximal minors. This is a polynomial identity in the universal coefficient ring. It proves the assertion before taking a radical, localizing at a tangent minor, or passing to residue fields. $\square$

The same calculation applies to an augmented locally free algebra $\mathcal{O} \oplus V \oplus S_2$ over a base scheme, with locally free layers, $S_2(V \oplus S_2) = 0$, and multiplication γ . Formula (8.2) therefore describes families of quadratic relations as well as families of generating planes.

8.2. The complete class of Hilbert function (1, 2, 2). Every local algebra in this class has $\mathfrak{m}^3 = 0$. The kernel of $\gamma : \operatorname{Sym}^2 V \rightarrow S_2$ is a line of binary quadratic forms. Over $\mathbb{C}$ its generator is either a square or a product of two independent linear forms. Thus the complete list of isomorphism types is

$$\begin{aligned} B_{\text{sq}} &= \mathbb{C}[x, y] / ((x, y)^3, y^2), \\ B_{\text{tf}} &= \mathbb{C}[x, y] / ((x, y)^3, xy). \end{aligned} \tag{8.3}$$

Indeed a linear splitting of $\mathfrak{m} \rightarrow V$ identifies B with $\mathbb{C} \oplus V \oplus S_2$, with multiplication determined entirely by γ ; there are no further extension coefficients when $\mathfrak{m}^3 = 0$.

In both cases $X = \operatorname{Gr}(2, \mathfrak{m})$ parametrizes unital three-planes. On its frame bundle put

$$M = \begin{pmatrix} a & c \\ b & d \end{pmatrix}, \quad \delta = ad - bc, \quad R = \mathbb{C}[a, b, c, d], \quad P = (a, c), \quad Q = (b, d), \quad \mathfrak{n} = (a, b, c, d).$$

The four entries of H are independent additional variables. All the following ideals are extended to $R[H]$ and restricted to the open where the stacked matrix (8.1) has rank two. A component absent from an individual affine chart is simply omitted there; the decomposition is irredundant on the whole frame bundle.

Theorem 8.2 (Complete primary decomposition in length five). *For every $m \geq 2$ the full multiplication failure scheme has the following primary decompositions.*

For B_{sq} ,

$$I_{\text{sq}} = \delta^2 P^2 = (\delta^2) \cap P^4, \quad \operatorname{Ass}(R/I_{\text{sq}}) = \{(\delta), P\}. \tag{8.4}$$

The prime P is embedded. More generally, for every $j \geq 1$,

$$I_{\text{sq}}^j = (\delta^{2j}) \cap P^{4j}. \tag{8.5}$$

For B_{tf} , set

$$\begin{aligned} J &= (ab, ad + bc, cd), \\ Q_0 &= J + P^2 + Q^2, \\ Q_* &= \delta^2 Q_0 + \mathfrak{n}^7. \end{aligned} \tag{8.6}$$

Then Q_0 and Q_* are $\mathfrak{n}$ -primary and

$$\begin{aligned} I_{\text{tf}} &= \delta^2 J = (\delta^2) \cap P^3 \cap Q^3 \cap Q_*, \\ \operatorname{Ass}(R/I_{\text{tf}}) &= \{(\delta), P, Q, \mathfrak{n}\}. \end{aligned} \tag{8.7}$$

The last three primes are embedded. The displayed Q_* is a specified primary representative, not a canonical choice of an embedded primary component.

These identities describe the entire schemes on X , including the extreme-corank locus T . Intrinsically, an embedded surface is indexed by a linear factor $L \subset V$ of the quadratic relation: its support is

$$Y_L = \{K \subset \mathfrak{m} : K \rightarrow V/L \text{ is zero}\} = \text{Gr}(2, S_2 + L) \simeq \mathbb{P}^2. \quad (8.8)$$

There is one such surface for a square and two for distinct factors. In the two-factor case there is also the embedded point $K = S_2$. There are no further associated strata.

The reduced support in both cases is the same integral normal divisor

$$\Delta = \{\text{rank}(K \rightarrow V) \leq 1\} \subset X.$$

Its singular locus is the point $K = S_2$. The generic multiplicity of D along Δ is two, whereas the nilradical of $\mathcal{O}_D$ has exact nilpotency index four. Neither full failure scheme is Cartier.

Proof. Choose bases in the quadratic layer, and rescale the middle symmetric basis vector where necessary. The two quadratic product matrices are

$$L_{\text{sq}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ ab & ad + bc & cd \end{pmatrix}, \quad L_{\text{tf}} = \begin{pmatrix} a^2 & 2ac & c^2 \\ b^2 & 2bd & d^2 \end{pmatrix}.$$

For the first matrix the target basis may be taken to be $x^2, 2xy$; for the second it is x^2, y^2 . Their two-minor ideals are, respectively, δP^2 and δJ . Theorem 8.1 gives the asserted full failure ideals. Notice the extra factor δ supplied by the linear layer; dropping that layer would produce a different primary problem.

For the square case, the P -adic associated graded ring of R is $\mathbb{C}[b, d][a, c]$, a domain. The P -adic order of δ is one. Consequently

$$(P^t : \delta^u) = P^{\max\{t-u, 0\}} \quad (t, u \geq 0).$$

It follows that $(\delta^2) \cap P^4 = \delta^2 P^2$, and the same calculation with $2j, 4j$ proves (8.5). The polynomial δ is prime, and powers of P are primary. The two components are irredundant: $\delta^2 \notin P^4$ and $a^4 \notin (\delta^2)$. This proves the associated-prime assertion.

For the second case we first prove

$$J = P \cap Q \cap Q_0. \quad (8.9)$$

Since $P \cap Q = PQ$, its quotient by J is spanned by the class of δ : the relations set $ab = cd = 0$ and $ad = -bc$. Every element of $\mathfrak{n}$ annihilates this class, and the class is nonzero by linear independence in degree two. In R/Q_0 a basis is

$$1, a, b, c, d, ad,$$

with $bc = -ad$ and all other quadratic monomials zero. Thus $\mathfrak{n}^3 \subset Q_0$, Q_0 is $\mathfrak{n}$ -primary, and $\delta = 2ad$ remains nonzero in this quotient. The natural map $PQ/J \rightarrow R/Q_0$ is therefore injective, proving (8.9). It is irredundant; the embedded prime $\mathfrak{n}$ is also witnessed directly by $\text{ann}_{R/J}(\bar{\delta}) = \mathfrak{n}$.

The same associated-graded argument as before gives

$$(P^3 : \delta^2) = P, \quad (Q^3 : \delta^2) = Q.$$

We claim that $(Q_* : \delta^2) = Q_0$. For a homogeneous polynomial of degree j , membership of its product with δ^2 in $\mathfrak{n}^7$ is equivalent to $j \geq 3$. More generally the homogeneous decomposition, and injectivity of multiplication by δ^2 in R , give

$$(\delta^2 Q_0 + \mathfrak{n}^7 : \delta^2) = Q_0 + \mathfrak{n}^3 = Q_0.$$

Intersecting with (δ^2) and using (8.9) proves (8.7). Since $Q_* \subset \mathfrak{n}$ and contains $\mathfrak{n}^7$, it is $\mathfrak{n}$ -primary. For completeness multiplication by δ^2 gives an injection $R/J \hookrightarrow R/I_{\text{tf}}$. Hence its three associated

primes $P, Q, \mathfrak{n}$ remain associated. The prime (δ) is the unique minimal prime of I_{tf} . This proves irredundance and excludes additional associated primes.

All these decompositions survive polynomial extension by H and localization. Each associated stratum actually meets the frame open: one may set H equal to the identity matrix. Descent is not a specialization assertion. The ideals P, Q express the intrinsic conditions (8.8); their sum expresses $K = S_2$. The determinant ideal is intrinsic, and the residual ideal $J = I_{\text{tf}} : (\delta^2)$ is invariant under a change of frame. Thus Q_0 and Q_* can equally be formed from these invariant ideal sheaves. The factor surfaces may be interchanged by an automorphism of B , but their unordered collection and the decomposition descend. The frame morphism is a smooth faithfully flat torsor, with no new associated primes introduced by its regular fibres. Alternatively, all assertions follow on the standard Grassmannian charts by solving for a frame minor, leaving exactly the displayed extended ideals. This establishes the claims on the whole of X .

The reduced equation on the frame cover is $\delta = 0$. It is an integral hypersurface whose singular locus is $M = 0$, of codimension three within that hypersurface. A hypersurface is Cohen–Macaulay; regularity in codimension one and Serre’s criterion give normality. Integrality on X also follows from the surjection $\text{Gr}_{\mathbb{P}(V)}(2, S_2 + \mathcal{L}) \rightarrow \Delta$ from an integral scheme. At a general point of Δ the residual ideal P^2 or J is a unit ideal, so the generic multiplicity is two.

In either case write $I = \delta^2 J'$, where $J' = P^2$ or J . One has $\delta \notin J'$ and $\delta^2 \in J'$; in the latter case use $\delta^2 = (ad + bc)^2 - 4abcd$. Therefore $\delta^4 \in I$ and $\delta^3 \notin I$. Since $\sqrt{I} = (\delta)$, the nilradical has exact index four. Nonvanishing is retained on the chart where H is invertible. Finally a nonzero principal ideal in the regular scheme X cannot have these embedded associated primes. Thus D is not Cartier. $\square$

8.3. Conductor equations on the embedded strata. The preceding theorem computes local rings across T , not just the fibres of the incidence over T . We now locate the conductor strata inside these primary supports. This also specifies which embedded components lie entirely in the extreme-corank locus.

Theorem 8.3 (Conductor position of the primary strata). *On the exact rank-one locus of $K \rightarrow V$, write $L = \text{im}(K \rightarrow V)$ and $H_0 = K \cap S_2$. Both are line bundles. The subalgebra locus T there is precisely the zero scheme of*

$$\gamma|_{\text{Sym}^2 L} : \text{Sym}^2 L \longrightarrow S_2/H_0. \quad (8.10)$$

On that scheme the conductor is the actual kernel

$$\text{cond}_B(W) = \ker(K \longrightarrow \text{Hom}(V, S_2/H_0)), \quad k \longmapsto (v \longmapsto \gamma(\bar{k}, v)). \quad (8.11)$$

If the induced map $L \rightarrow \text{Hom}(V, S_2/H_0)$ has exact rank zero, the conductor is K and the conductor action has rank one. If it has exact rank one, the conductor is H_0 and the action has rank two. These kernel statements commute with base change on the specified scheme-theoretic rank strata.

For B_{sq} the embedded surface $Y_{\langle y \rangle}$ is contained in T . On its rank-one part the action rank is one exactly when $H_0 = \langle xy \rangle$, and is two otherwise. For B_{tf} the two embedded surfaces are $Y_{\langle x \rangle}$ and $Y_{\langle y \rangle}$. Their rank-one intersections with T are, respectively, $H_0 = \langle x^2 \rangle$ and $H_0 = \langle y^2 \rangle$; the conductor action has rank one on both intersections. Elsewhere on the rank-one part of T it has rank two.

At $K = S_2$, in both types, $W = \mathbb{C} \oplus S_2$ has conductor S_2 and quotient $B/S_2 = \mathbb{C} \oplus V$ with square-zero augmentation ideal. The incidence fibre there is $\mathbb{P}(V)$, whereas the action-rank-two fibres above are spectra of the dual numbers. The extra embedded point at $K = S_2$ occurs precisely in the two-factor type. Thus numerical action rank alone does not determine associated primes: the quadratic relation in the conductor data is indispensable.

Proof. Since $S_2 \mathfrak{m} = 0$, one has $W^2 = \mathcal{O} \cdot 1 + K + \gamma(L^2)$. The quotient S_2/H_0 is a line bundle, so closure under multiplication is exactly (8.10), including its nonreduced zero-scheme structure.

An element $\lambda \cdot 1 + k \in W$ in the conductor must have $\lambda = 0$: multiplication by S_2 and projection to the free line S_2/H_0 is scalar multiplication by λ . For $k \in K$, multiplication by S_2 is zero, and multiplication by a lift of $v \in V$ is $\gamma(\bar{k}, v)$. It belongs to W exactly when it belongs to H_0 . This proves (8.11) over every coefficient ring on the indicated rank locus. Its restriction to the exact-rank strata is a kernel of constant rank, so the sequences split locally and pull back. The conductor has rank two or one accordingly; as $\text{rank } W = 3$, the action ranks are one or two.

In the square type, $\gamma(y^2) = 0$ and $\gamma(y, V) = \langle xy \rangle$. This proves the assertion about the entire embedded surface and its action-rank jump. If $L \neq \langle y \rangle$, then $\gamma(L, V) = S_2$, so action rank one cannot occur on its rank-one subalgebra locus. In the two-factor type, multiplication by x has image $\langle x^2 \rangle$ and multiplication by y has image $\langle y^2 \rangle$. For any other line both quadratic basis vectors occur and the image is S_2 . These observations prove all the rank-one assertions. They are equations in the universal line and quotient bundles, rather than an appeal to continuity of ranks.

At $K = S_2$ the conductor computation is immediate. The action-rank-one incidence is the scheme of length-two quotients of $\mathbb{C} \oplus V$, hence the projective line: its binary cubic square obstruction vanishes identically. On the action-rank-two locus the rank-two conductor algebra is local with square-zero augmentation line, since every quadratic product in K lies in H_0 . Proposition 7.2 therefore gives the spectrum of the dual numbers on geometric fibres. The assertion about the embedded point follows from Theorem 8.2, not from the incidence fibre list. $\square$

On the rank-one, action-rank-one loci just described, the residual rank-three algebra is curvilinear: the surviving tangent vector has nonzero square. Thus its length-two quotient scheme has length three. The change from these cubic fibres, or the quadratic fibres, to the projective line at $K = S_2$ is accompanied by the local equations (8.4) and (8.7). Those equations, rather than the dimensions of the fibres, determine the nilpotent boundary.

Proposition 8.4 (Collision of the quadratic factors). *The algebras*

$$\mathcal{B}_\tau = \mathbb{C}[\tau, x, y] / ((x, y)^3, y^2 - \tau xy)$$

form a finite free rank-five family over $\mathbb{C}[\tau]$. On the full unital three-plane frame space its failure ideal in every degree $m \geq 2$ is

$$\delta^2(a(a + \tau b), 2ac + \tau(ad + bc), c(c + \tau d)). \quad (8.12)$$

Its fibre at $\tau = 0$ is the square-type scheme, and every fibre at $\tau \neq 0$ is the two-factor-type scheme. Thus the primary strata in Theorem 8.2 occur in one family of full Fitting schemes with fixed algebra length and Hilbert function. No flatness of that failure family, or compatibility of a chosen primary decomposition with specialization, is asserted.

Proof. The elements $1, x, y, x^2, xy$ are a free basis over $\mathbb{C}[\tau]$; the displayed relation is the only elimination in degree two and all terms of degree at least three vanish. The quadratic matrix, in the product basis without rescaling the middle column, is

$$\begin{pmatrix} a^2 & ac & c^2 \\ 2ab + \tau b^2 & ad + bc + \tau bd & 2cd + \tau d^2 \end{pmatrix}.$$

Its minors are δ times the three displayed generators. Theorem 8.1 proves the ideal identity over $\mathbb{C}[\tau]$ and after specialization. The quadratic relation is $y(y - \tau x)$, proving the assertions about the fibres. $\square$

Corollary 8.5 (Projective realization of the complete boundary). *Let $Z \subset \mathbb{P}^2$ have homogeneous ideal $((X, Y)^3, Y^2)$ or $((X, Y)^3, XY)$ in $\mathbb{C}[T, X, Y]$. For every $n \geq 5$, the inverse-image linear series restricting to all generating three-planes on Z have, in each symmetric degree $m \geq 2$, exactly*

the primary schemes of Theorem 8.2, on their smooth unit covers. In particular the embedded conductor-boundary components are components of actual global section-multiplication failure.

Proof. The ideals are saturated because T is a nonzerodivisor on the quotient. The affine algebras have top graded degree two, so their homogeneous ideals have regularity three. Apply Theorem 10.1, using $n \geq 2 \cdot 3 - 1$, and trivialize $\mathcal{O}_Z(n)$ by T^n . The cokernel isomorphism transfers the full Fitting ideal. Finally Remark 7.3 transfers the equations from unital planes to a smooth cover of the unrestricted generating Grassmannian. No boundary is removed in either step. $\square$

9. EMBEDDED PRIMARY STRATA FOR CONNECTED LOCAL ALGEBRAS

The incidence of Theorem 7.1 transfers more than the support of a quotient scheme. On its quadratic-generation open it transfers its local rings, and hence its primary structure. We apply it to a connected noncurvilinear family in which the full multiplication failure scheme has an embedded component.

For $h \geq 5$ set

$$B_h = \mathbb{C}[x, y]/(x^2, xy, y^h), \quad \dim_{\mathbb{C}} B_h = h + 1. \quad (9.1)$$

Let $F_0 = 0$, $F_1 = 1$, and $F_{j+1} = uF_j + vF_{j-1}$, and put

$$\begin{aligned} J_h &= (F_h, vF_{h-1}) \subset \mathbb{C}[u, v], \\ I_h &= (F_h, vF_{h-1}, bv, b^2u) \subset \mathbb{C}[b, u, v], \\ P_h &= (u, v), \quad Q_h = (b^2, bv, F_h, vF_{h-1}). \end{aligned} \quad (9.2)$$

The equality $J_h = (F_h, F_{h+1})$ follows from the recurrence.

Theorem 9.1 (A connected embedded-primary family). *Consider the unital Grassmannian of $(h-1)$ -planes in B_h , and its full zeroth-Fitting multiplication failure scheme D_h in any degree $m \geq 3$. At the plane*

$$W_0 = \langle 1, x, y^2, y^3, y^5, \dots, y^{h-1} \rangle \quad (\text{the term } y^4 \text{ is omitted}), \quad (9.3)$$

there is a Zariski neighbourhood in D_h isomorphic to

$$\text{Spec}((\mathbb{C}[b, u, v]/I_h)[t_1, \dots, t_{h-2}])_f, \quad f(0) = 1. \quad (9.4)$$

Here the variables t_i are actual Grassmannian coordinates, and f is the coefficient of a specified quadratic product in the one-dimensional quotient of the conductor hyperplane by W . This is a local description of the full failure scheme, not of a transverse test family.

The ideal in (9.2) has the irredundant primary decomposition

$$I_h = P_h \cap Q_h. \quad (9.5)$$

Its associated primes are (u, v) and (b, u, v) . The first is the minimal prime and the second is embedded. The same decomposition, with the ideals extended and localized, describes (9.4). The reduced support has codimension $h - 3 \geq 2$ in the unital Grassmannian, so this failure scheme is not Cartier. Its nilradical has exact nilpotency index $2h - 3$. For the displayed choice of embedded primary ideal,

$$\dim_{\mathbb{C}} \mathbb{C}[b, u, v]/Q_h = \binom{h}{2} + h - 1, \quad \dim_{\mathbb{C}} \sqrt{I_h}/I_h = \binom{h}{2} + h - 3. \quad (9.6)$$

The second formula concerns the coefficient ring before adjoining the Grassmannian variables. It records a nilradical supported at the embedded point, rather than multiplicity along the generic point of a Cartier divisor.

Proof. We first compute an entire open chart of $\text{Hilb}^2(B_h)$. Choose the chart on which $1, y$ is a basis of the universal quotient. Write $y^2 = uy + v$ and $x = a + by$. The relation $xy = 0$ gives $a = -bu$ and $bv = 0$. After this substitution, $x^2 = 0$ is equivalent to $b^2u = 0$: its constant equation follows from $bv = 0$ and $b^2u = 0$. Finally

$$y^h = F_h y + v F_{h-1}$$

shows that $y^h = 0$ is equivalent to the two generators of J_h . These calculations give a quotient algebra free on $1, y$ over $\mathbb{C}[b, u, v]/I_h$, and conversely every such quotient has these coefficients. They prove an isomorphism of the chart functor with $\text{Spec } \mathbb{C}[b, u, v]/I_h$ over arbitrary coefficient rings. No radical has been taken.

Here is a direct primary calculation. Write $A_h = \mathbb{C}[u, v]/J_h$ and $\mathfrak{n} = (u, v)A_h$. It is an Artin local algebra, with residue field $\mathbb{C}$. In $A_h[b]$, coefficient comparison in the variable b gives

$$\mathfrak{n}A_h[b] \cap (b^2, bv) = (bv, b^2\mathfrak{n}) = (bv, b^2u).$$

Lifting to $\mathbb{C}[b, u, v]$ proves (9.5). The ideal P_h is prime; the quotient by Q_h is an Artin local ring supported at (b, u, v) , so Q_h is primary. The decomposition is irredundant: $b^2 \in Q_h \setminus P_h$, whereas $u \in P_h \setminus Q_h$. The asserted associated primes follow. Polynomial extension and localization at a function nonzero at the specified point retain both components.

For clarity, the numerical invariants can be read without any primary-decomposition algorithm. The ring A_h is the contact algebra G_h of Theorem 5.3. It has dimension $\binom{h}{2}$, and

$$A_h/(v) \simeq \mathbb{C}[u]/(u^{h-1}), \quad \mathfrak{n}^{2h-3} = 0, \quad u^{2h-4} \neq 0.$$

These last assertions also follow from its two-row Schur basis: $\deg u = 1$, $\deg v = 2$, the top weighted degree is $2h - 4$, and the coefficient of the top Schur class in u^{2h-4} is nonzero. This is the classical basis input recorded in [31]. As vector spaces the quotient by Q_h is $A_h \oplus b(A_h/(v))$, proving the first length formula. In the quotient by I_h , the nilradical is

$$\mathfrak{n} \oplus b(u\mathbb{C}[u]/(u^{h-1})).$$

All coefficients of b^j for $j \geq 2$ are scalars. This proves the second length formula. The inclusion of the constant coefficient ring A_h is injective, so the upper bound $2h - 3$ on nilpotency is attained. Localization at $f(0) = 1$ does not kill its nonzero top class: it is supported at the retained maximal ideal.

It remains to prove that this chart is realized by the full multigenerator failure scheme. At $b = u = v = 0$, the conductor hyperplane is

$$E_0 = \langle 1, x, y^2, \dots, y^{h-1} \rangle.$$

The plane W_0 in (9.3) is a hyperplane in E_0 , and $W_0^2 = E_0$, since $(y^2)^2 = y^4$. In the universal quotient chart the conductor hyperplane has the basis

$$1, \quad x - by, \quad y^j - F_j y \quad (2 \leq j \leq h-1).$$

Take the Grassmannian chart in which the vector $y^4 - F_4 y$ is the omitted direction, and add its multiples t_i to each of the other nonconstant basis vectors. This gives $h - 2$ free coordinates on $\text{Gr}(h - 2, E/\mathcal{O})$. Let q_2 be the resulting lift of the y^2 basis vector. Its square belongs to E , because E is an algebra. The coefficient f of q_2^2 in E/W , in the omitted basis direction, equals 1 at the origin. On $f \neq 0$ one has $W^2 = E$. Theorem 7.1 now identifies this localized polynomial chart with an open neighbourhood of W_0 in D_h , proving (9.4).

The unit Grassmannian has dimension $2(h - 2)$, whereas the support of (9.4) has dimension $1 + (h - 2) = h - 1$. Their difference is $h - 3$. This proves the non-Cartier assertion and finishes the proof. $\square$

For $h = 5$ the theorem gives four-planes in a connected length-six algebra. Its coefficient ideal is explicitly

$$(u^4 + 3u^2v + v^2, u^3v + 2uv^2, bv, b^2u).$$

The reduced support is the b -axis, the chosen embedded primary quotient has length 14, and the nilradical has dimension 12 and index 7. Increasing h changes both the higher-jet equations and the embedded nilpotent algebra; it does not merely change the exponent of a fixed determinant divisor. The parameter b couples the extra square-zero tangent direction x to the higher contact coefficients u, v through bv and b^2u .

Corollary 9.2 (A three-generator member after adjoining a reduced factor). *For every $s \geq 1$, let $B = (\mathbb{C}[x, y]/(x^2, y^2)) \times \mathbb{C}^s$. The full stable failure scheme of unital codimension-two planes has a chart, meeting an embedded associated stratum, of the form*

$$\text{Spec}((\mathbb{C}[a, b]/(ab, b^2))[t_1, \dots, t_{s+1}])_g, \quad g(0) = 1.$$

Its primary decomposition is the extension of $(ab, b^2) = (b) \cap (a, b)^2$. In particular $s = 1$ gives three multiplication generators and an embedded component on the full failure scheme, rather than on a selected slice.

Proof. Use the open-and-closed part of $\text{Hilb}^2(B)$ supported in the first factor. Proposition 5.5 gives the coefficient chart $\mathbb{C}[a, b]/(ab, b^2)$ there. At its origin the conductor hyperplane is $\langle 1, y, xy \rangle \times \mathbb{C}^s$. Let e_i denote the standard idempotent of the i th reduced factor. The plane generated by

$$1_B, \quad (y, e_1), \quad (xy, 0), \quad e_2, \dots, e_s$$

has rank $s + 2 = \dim B - 2$. Its square contains $e_1 = (y, e_1)^2$, so its quadratic image is the entire conductor hyperplane. The relative Grassmannian chart has $s + 1$ parameters. Localize at the quadratic-generation minor nonzero at this point and apply Theorem 7.1. The displayed primary identity is retained under this polynomial extension and localization. $\square$

The connected family of Theorem 9.1 and the three-generator corollary serve different purposes. The first gives unbounded connected local complexity; the second supplies a three-generator instance of the same incidence mechanism. Both also describe smooth local covers of the unrestricted generating Grassmannians by Remark 7.3. Neither uses a claim that primary decomposition survives arbitrary specialization.

10. CONDUCTOR TRANSPORT FOR ARBITRARY FINITE PROJECTIVE SCHEMES

The finite-algebra results apply to arbitrary zero-dimensional projective schemes once the conductor kernel is filled. This can be done by a normal-layer argument in which Castelnuovo–Mumford regularity replaces the monomial calculation for a homogeneous fat point. We distinguish the regularity of a homogeneous ideal from that of its associated sheaf; the distinction is essential for its ordinary square.

Theorem 10.1 (Regularity-controlled conductor transport). *Let $Z \subset \mathbb{P}_{\mathbb{C}}^e$, $e \geq 1$, be a nonempty zero-dimensional closed subscheme, and let $I \subset R = \mathbb{C}[X_0, \dots, X_e]$ be its saturated homogeneous ideal. Put*

$$r = \text{reg } I, \quad s = \text{reg}(I^2),$$

where I^2 is the ordinary homogeneous ideal square. If

$$n \geq \max\{r, s - 1\}, \tag{10.1}$$

then every generating subspace $A \subset H^0(Z, \mathcal{O}_Z(n))$, with $U = \text{res}_Z^{-1} A$ and $K_t = H^0(\mathbb{P}^e, \mathcal{I}_Z(t))$, satisfies

$$K_n U^{m-1} = K_{mn} \quad (m \geq 2). \tag{10.2}$$

Consequently restriction induces a canonical isomorphism

$$\operatorname{coker}(\operatorname{Sym}^m \mathcal{U} \rightarrow H^0(\mathbb{P}^e, \mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \operatorname{coker}(\operatorname{Sym}^m \mathcal{A} \rightarrow H^0(Z, \mathcal{O}_Z(mn)) \otimes \mathcal{O}) \quad (10.3)$$

on the entire generating Grassmannian, for every $m \geq 2$. In particular all Fitting ideals agree.

The simpler sufficient range

$$n \geq 2 \operatorname{reg} I - 1 \quad (10.4)$$

always implies (10.1). It has no restriction on the local embedding dimension, the support partition, or a monomial presentation of Z .

Proof. We use the following classical regularity facts. An ideal of regularity r is generated in degrees at most r . If an ideal has regularity s , then it agrees with its saturation in degrees at least s , and the first cohomology of its associated sheaf vanishes in degrees at least $s - 1$; see [35, Theorem 1.3]. Moreover

$$\operatorname{reg}(I^2) \leq 2 \operatorname{reg} I \quad (10.5)$$

for a zero-dimensional projective scheme, by [35, Theorem 1.8]. This is a bound for the ordinary ideal square, not merely its sheafification. It proves the implication from (10.4). We prove the sharper assertion under (10.1).

Choose $g \in U$ whose restriction generates $\mathcal{O}_Z(n)$ as an invertible $\mathcal{O}_Z$ -module. Such a section exists by Lemma 3.1. Let $t \geq n$. Since $t \geq s - 1$, the conormal sequence gives a surjection

$$K_t \longrightarrow H^0(Z, (\mathcal{I}_Z/\mathcal{I}_Z^2)(t)). \quad (10.6)$$

Multiplication by g is an isomorphism between the conormal twists t and $t + n$, since its restriction is a unit after trivializing $\mathcal{O}_Z(n)$. Thus, for any $f \in K_{t+n}$, one can choose $q \in K_t$ for which $f - gq$ lies in $H^0(\mathbb{P}^e, \mathcal{I}_Z^2(t + n))$.

The ideal I is generated in degrees at most r , and $n, t \geq r$. Factoring coefficient monomials between two chosen ideal generators therefore gives the elementary homogeneous equality

$$(I^2)_{t+n} = I_n I_t. \quad (10.7)$$

Indeed for generators of degrees $a, b \leq r$, a coefficient monomial of degree $t + n - a - b$ can be split into factors of degrees $n - a$ and $t - b$. This proves both containments. Since $s \geq 2$ and $n \geq s - 1$, one has $t + n \geq s$. Agreement with saturation in that degree identifies the preceding sheaf section space with $(I^2)_{t+n}$. Also $K_t = I_t$, because I is saturated. We have proved

$$K_{t+n} = gK_t + K_n K_t = U K_t. \quad (10.8)$$

The first equality means equality of the sum with the whole kernel; the last follows from $g, K_n \subset U$, with the reverse inclusion clear from vanishing on Z . Iterating for $t = n, 2n, \dots$ proves (10.2).

Restriction in degrees n and mn is surjective by regularity. The restriction sequence $0 \rightarrow K_n \rightarrow U \rightarrow A \rightarrow 0$ is locally split on the parameter Grassmannian. The kernel on symmetric powers is generated by $K_n \otimes \operatorname{Sym}^{m-1} U$, whose multiplication image is the full target restriction kernel by (10.2). Taking cokernels of the restriction square proves (10.3). Its map is restriction itself, so the local choices of g disappear from the result. The relative version below supplies, in particular, the asserted scheme-level statement over this whole Grassmannian. $\square$

Theorem 10.2 (Relative finite schemes and base change). *Let S be a locally Noetherian $\mathbb{C}$ -scheme and $\mathcal{Z} \subset \mathbb{P}_S^e$ a finite locally free family. Suppose the saturated homogeneous ideal I_z of every geometric fibre has regularity at most r . For $n \geq 2r - 1$, the kernel identity and cokernel comparison of Theorem 10.1 hold on each relative generating Grassmannian over S and commute with arbitrary base change. The same conclusion holds under uniform bounds $\operatorname{reg} I_z \leq r$, $\operatorname{reg}(I_z^2) \leq s$ and $n \geq \max\{r, s - 1\}$.*

No hypothesis that formation of a conductor ideal or of a primary decomposition commutes with specialization is used. The assertion is about the actual coherent multiplication cokernels.

Proof. Let $p : \mathbb{P}_S^e \rightarrow S$. For $t \geq n$, regularity on geometric fibres and cohomology and base change give split exact sequences locally on S ,

$$0 \longrightarrow p_* \mathcal{I}_{\mathcal{Z}}(t) \longrightarrow p_* \mathcal{O}(t) \longrightarrow p_* \mathcal{O}_{\mathcal{Z}}(t) \longrightarrow 0.$$

Their terms are vector bundles and their formation commutes with base change. One may also obtain this directly from fibrewise surjectivity of restriction, the local freeness of its two outer bundles, and Nakayama's lemma. The resulting kernel has the required fibre by the exact sequence on each fibre.

Over the relative Grassmannian, form the multiplication morphism

$$p_* \mathcal{I}_{\mathcal{Z}}(n) \otimes \mathrm{Sym}^{m-1} \mathcal{U} \longrightarrow p_* \mathcal{I}_{\mathcal{Z}}(mn).$$

It is surjective on every geometric fibre by the preceding theorem, using (10.5) if only the bound r is specified. Its coherent cokernel therefore vanishes by Nakayama's lemma. The same symmetric-power restriction square proves the relative cokernel isomorphism. The restriction bundles and their split sequences pull back, and a surjective morphism remains surjective after every pullback. Hence the construction proves arbitrary base change, not just flat base change. Fitting ideals commute with base change for finitely presented modules. This argument does not require flatness of the first infinitesimal neighbourhood of $\mathcal{Z}$ or base change for a selected primary component. $\square$

The bound recovers the fat-point range $n \geq 2h - 1$, since the homogeneous ideal of an order- h fat point has regularity h . The direct triangular proof in Theorem 12.2 is retained, as is its sharp boundary example. The present proof also applies to nonmonomial local schemes, unions with unequal local lengths, and finite flat deformations with a uniform regularity bound. A uniform bound is available on a fixed projective Hilbert scheme by boundedness; sharpness of $2r - 1$ is not asserted for each individual finite scheme or each generating rank.

Corollary 10.3 (Global embedded multiplication failure). *Let $h \geq 5$ and let $Z_h \subset \mathbb{P}^2$ have saturated ideal*

$$I = (X^2, XY, Y^h) \subset \mathbb{C}[T, X, Y].$$

For every $n \geq 2h - 1$, take all inverse-image series of generating $(h - 1)$ -planes on Z_h . In every degree $m \geq 3$, their global multiplication failure scheme has, on a smooth unit cover near the plane (9.3), the local primary description (9.4)–(9.5). It has the embedded component and exact nilradical index stated there. The coherent cokernel comparison persists in any finite flat family with the stated regularity bound.

Proof. The variable T is a nonzerodivisor on $\mathbb{C}[T, X, Y]/I$, so I is saturated and the scheme is supported at $[1 : 0 : 0]$. Its affine algebra is B_h . The standard monomials in $\mathbb{C}[X, Y]/I$ are $1, X, Y, \dots, Y^{h-1}$, and its top degree is $h - 1$; thus $\mathrm{reg} I = h$. For comparison, the minimal monomial generators of I^2 are

$$X^4, X^3Y, X^2Y^2, XY^{h+1}, Y^{2h},$$

and its top quotient degree is $2h - 1$, so $\mathrm{reg}(I^2) = 2h$. The conductor theorem applies at $n \geq 2h - 1$. Trivialize $\mathcal{O}_{Z_h}(n)$ by T^n and apply Theorem 9.1 and the unit-cover construction. The equality of coherent cokernels gives the same Fitting scheme globally. The final assertion is the relative theorem, not a claim that the displayed embedded primary ideal persists unchanged under every deformation. $\square$

10.1. Cohomology and arbitrary base change in conductor transport. We make explicit the bundle assertions used in Theorem 10.2. Write $p : \mathbb{P}_S^e \rightarrow S$ and $q : \mathcal{Z} \rightarrow S$. The sheaf $\mathcal{I}_{\mathcal{Z}}$ is S -flat: it is the kernel of a surjection between S -flat sheaves $\mathcal{O}$ and $\mathcal{O}_{\mathcal{Z}}$. For $t \geq \max\{0, r - 1\}$ the two modules

$$E_t = p_* \mathcal{O}(t), \quad F_t = q_* \mathcal{O}_{\mathcal{Z}}(t)$$

are locally free and commute with arbitrary base change. The first is the familiar homogeneous polynomial bundle; for the second use that q is finite locally free and $\mathcal{O}_{\mathcal{Z}}(t)$ is invertible. The fibre regularity bound gives $H^1(\mathbb{P}_s^e, \mathcal{I}_{\mathcal{Z}_s}(t)) = 0$ in this range. Hence the restriction map $E_t \rightarrow F_t$ is surjective on each geometric fibre and, by Nakayama's lemma, on S .

Its kernel K_t is locally free because the quotient F_t is locally free, so

$$0 \longrightarrow K_t \longrightarrow E_t \longrightarrow F_t \longrightarrow 0$$

is locally split. The long exact sequence of direct images identifies $K_t = p_*\mathcal{I}_{\mathcal{Z}}(t)$ and gives $R^1p_*\mathcal{I}_{\mathcal{Z}}(t) = 0$. In fact all higher direct images vanish in this range: $\mathcal{Z}$ is finite over S , and $\mathcal{O}(t)$ on projective space has no higher direct images for $t \geq 0$.

After any base change $S' \rightarrow S$ the displayed split sequence stays exact. The corresponding sequence of sheaves on projective space also stays exact because the original family is flat. Pushforward identifies the new kernel with $p'_*\mathcal{I}_{\mathcal{Z}'}(t)$; restriction is still surjective and its first higher direct image is still zero. This proves the arbitrary base-change assertion directly, without assuming S' reduced, flat over S , or Noetherian.

Now form the relative generating Grassmannian and let $\mathcal{U}$ be the inverse image of its tautological restricted series. For $m \geq 2$ the map

$$K_n \otimes \mathrm{Sym}^{m-1}\mathcal{U} \longrightarrow K_{mn}$$

is a morphism of locally free sheaves. If the uniform bounds are $\mathrm{reg} I_s \leq r$ and $\mathrm{reg}(I_s^2) \leq s$, the single-fibre conductor proof applies for $n \geq \max\{r, s-1\}$. If only r is given, Sidman's ordinary-square bound supplies $s = 2r$. These bounds are used on the geometric fibres only: no flatness or base change is required for $\mathcal{I}_{\mathcal{Z}}^2$ or its conormal quotient. Thus the displayed morphism is fibrewise surjective, and its finitely presented cokernel is zero. It remains surjective after every pullback.

Finally the locally split sequence $0 \rightarrow K_n \rightarrow \mathcal{U} \rightarrow \mathcal{A} \rightarrow 0$ shows that the kernel of $\mathrm{Sym}^m \mathcal{U} \rightarrow \mathrm{Sym}^m \mathcal{A}$ is generated by $K_n \otimes \mathrm{Sym}^{m-1} \mathcal{U}$. Its multiplication image is exactly the target restriction kernel K_{mn} . The induced isomorphism of multiplication cokernels is therefore canonical restriction and commutes with arbitrary base change. This explains all relative assertions without replacing a coherent cokernel by its fibre dimensions.

11. THREE GENERATORS ON A REDUCED CONTACT DIVISOR

This theorem realizes Haiman's diagonal ideal as the stable Fitting ideal of an actual multiplication map. The all-powers identity is Haiman's input, not a theorem deduced here from multiplication. The resulting global-section failure schemes and their coherent cokernels are obtained using the conductor comparison.

Theorem 11.1 (The complete stable three-plane ideal). *Let $D = p_1 + \cdots + p_d$ be reduced, $d \geq 4$, and let A vary on the entire generating open $\mathrm{Gr}(3, H^0(\mathcal{O}_D))^\circ$. On the smooth unit-frame cover write $A = \langle a, ax, ay \rangle$; thus the matrix with rows $(1, x_i, y_i)$ has rank three. For $m \geq d-1$, the zeroth Fitting ideal J_m is independent of m , and*

$$J_m = \bigcap_{i < j} P_{ij}, \quad P_{ij} = (x_i - x_j, y_i - y_j). \quad (11.1)$$

For every $q \geq 1$ one has the stronger identity

$$J_m^q = \bigcap_{i < j} P_{ij}^q. \quad (11.2)$$

These are complete primary decompositions, including all component intersections. Thus J_m is radical, all its components are smooth of codimension two on this cover and on the fixed- D Grassmannian, and neither it nor any of its powers has embedded associated points. The corank

is d minus the number of distinct pairs (x_i, y_i) . For global binary subseries the same statements hold for $n \geq 2d - 3$.

Proof. After dividing the target by a^m , the multiplication matrix has as columns all monomials in x, y of total degree at most m , evaluated at the d labelled points. Lemma 3.2 identifies, over the entire coefficient ring, the ideal of its d -minors for $m \geq d - 1$ with the ideal generated by determinants of arbitrary monomial columns. Those determinants span the alternating polynomials in the d pairs of variables: antisymmetrize a monomial and use linearity. Haiman's theorem [30, Corollary 3.8.3] states that the ideal generated by these alternants has every power equal to the intersection of the corresponding powers of the pair-diagonal ideals. It gives (11.1) and (11.2) before any localization; hence they hold on the rank-three open. In that corollary Haiman's n is our number d of labelled points, and Haiman's exponent d is our power q . These two indices are independent; no restriction on q is imposed. This use of the diagonal-ideal theorem is essential.

Each P_{ij} is a smooth prime ideal of codimension two; its powers are primary (use two normal coordinates in a regular local ring). The intersections therefore have exactly the indicated associated primes. Smooth faithful flat descent gives the fixed- D assertions. Polynomial functions separate any finite set of distinct affine points, so their image dimension is the number of distinct pairs. The truncation lemma gives the stated corank in this finite degree. Finally apply Theorem 2.2 with $k = 3$. $\square$

The stable moving scheme over the reduced-divisor stratum is integral, though it need not be smooth or unibranched. Its normalization marks one equal pair and is smooth and finite. Indeed the ordered configuration cover is etale and the permutation group is transitive on pairs. Upstairs the reduced scheme is the union in (11.1); its normalization is the disjoint union of the smooth pair diagonals. This construction descends. A point is smooth precisely when only one pair coincides; two or more failed pairs give distinct local branches after etale base change. No assertion about nonreduced divisors is inferred from this etale calculation; those have their own quotient-algebra structure in Section 5 when $k = d - 1$.

12. NONREDUCED MULTIGENERATOR SCHEMES AND FAT-POINT TRANSPORT

We next determine a multiplication failure scheme with arbitrarily large embedding dimension and nilpotence order. It is not a diagonal arrangement and its proof does not use Haiman's theorem. Fix integers $e \geq 2$ and $h \geq 3$, and put

$$B_{e,h} = \mathbb{C}[z_1, \dots, z_e]/(z_1, \dots, z_e)^h, \quad d = \binom{e+h-1}{e}, \quad N = \binom{e+h-1}{e+1}. \quad (12.1)$$

Write $\mathfrak{m}$ for its augmentation ideal. On $X = \text{Gr}(e+1, B_{e,h})^\circ$ the augmentation of the universal bundle $\mathcal{A}$ is surjective. Its kernel K has rank e . Let $\Delta \subset X$ be the determinant divisor of the canonical map

$$K \longrightarrow (\mathfrak{m}/\mathfrak{m}^2) \otimes \mathcal{O}_X. \quad (12.2)$$

Thus this divisor is defined without a choice of unit or coordinates.

Theorem 12.1 (Exact primary law for truncated local algebras). *For every $m \geq h - 1$, on the entire generating Grassmannian X ,*

$$\text{Fitt}_0 \text{ coker}(\text{Sym}^m \mathcal{A} \rightarrow B_{e,h}) = \mathcal{I}_\Delta^N. \quad (12.3)$$

The divisor Δ is nonempty, integral, and Cartier. The displayed ideal is primary, its exact nilradical index is N , and its generic transverse length is N . For every $q \geq 1$ its q th ordinary power has the complete primary description $\mathcal{I}_\Delta^{Nq}$; there are no embedded associated points. On a

smooth unit-frame cover, Δ is a generic e by e determinant, and its singular locus is given by matrix rank at most $e - 2$.

More precisely, choose a unit $a \in \mathcal{A}$ and write $\mathcal{A} = \langle a, ax_1, \dots, ax_e \rangle$, with $x_i \in \mathfrak{m}$. After multiplying the target by a^{-m} , the cokernel in (12.3) is the cokernel of the square substitution map

$$T_x : B_{e,h} \longrightarrow B_{e,h}, \quad z_i \longmapsto x_i.$$

If M is the matrix of linear coefficients of the x_i , then $\det T_x = (\det M)^N$. These presentation and ideal identities hold after arbitrary base change. The primary assertions refer to the universal parameter space X over $\mathbb{C}$, not to every specialization. For $m < h - 1$ the zeroth Fitting ideal is zero: the failure scheme is all of X , not the empty scheme. Thus the stated stable threshold is sharp for this family.

Proof. A unit exists locally because the augmentation of $\mathcal{A}$ is onto; subtracting constants from the remaining normalized basis vectors puts all x_i in $\mathfrak{m}$. The normalized multiplication matrix consists of the columns x^α with $|\alpha| \leq m$. Columns of total degree at least h vanish. When $m \geq h - 1$ the remaining columns are exactly the d columns of T_x ; all other columns are zero. This proves the cokernel assertion, over any coefficient algebra.

The map T_x preserves the $\mathfrak{m}$ -adic filtration and its block on $\mathfrak{m}^j/\mathfrak{m}^{j+1}$ is $\text{Sym}^j M$. Hence

$$\det T_x = \prod_{j=1}^{h-1} \det(\text{Sym}^j M) = (\det M)^{\sum_{j=1}^{h-1} \binom{e+j-1}{e}} = (\det M)^N.$$

For completeness, the middle identity follows by taking M diagonal: among all degree- j monomials each diagonal entry occurs with total exponent $\binom{j}{e} \binom{e+j-1}{e-1} = \binom{e+j-1}{e}$. The identity holds for arbitrary matrices because both sides are polynomial functions and diagonalizable matrices with distinct eigenvalues form a dense open over $\mathbb{C}$. It is consequently a polynomial identity and remains valid on nonreduced bases. No condition on the coefficients of degree at least two was used.

The determinant of (12.2) differs from $\det M$ by a unit: the vectors ax_i form a basis of K , and their linear terms are the linear terms of x_i multiplied by the nonzero augmentation of a . Thus the ideal equality descends canonically. To check its geometric structure, send A to $K = A \cap \mathfrak{m}$. This identifies X with the total space of the quotient bundle $\mathfrak{m}/K$ over $\text{Gr}(e, \mathfrak{m})$: a lift of the augmented line has the unique form $1 + v$ modulo K . The projection of its universal e -plane to $\mathfrak{m}/\mathfrak{m}^2$ vanishes in determinant precisely on the standard Schubert divisor

$$\Sigma = \{K \in \text{Gr}(e, \mathfrak{m}) : K \cap \mathfrak{m}^2 \neq 0\}.$$

This is the integral Schubert divisor for the fixed projection with kernel $\mathfrak{m}^2$, and Δ is its pullback along the quotient-bundle projection. This identifies the divisor globally. For an elementary verification of integrality, its frame cover has independent linear coefficients M and independent higher coefficients, subject only to full column rank of the combined matrix. The generic determinant is irreducible. Its zero scheme remains nonempty on the frame open, as witnessed by

$$x_1 = z_1, \dots, x_{e-1} = z_{e-1}, \quad x_e = z_1^2.$$

The standard integral determinant hypersurface on that open descends to Σ , and its vector-bundle pullback is integral and Cartier.

Locally its equation is a prime element in a regular local ring. Every positive power is primary; alternatively, a hypersurface in a regular scheme has no embedded associated points and its support here has a single generic point. The radical of (δ^N) is (δ) , and δ^{N-1} is nonzero modulo δ^N at its generic point. This proves the exact nilradical index and length, and the same argument proves the assertion for all powers. The partial derivatives of a determinant are its cofactors, giving the stated singular locus. Finally $\text{rank Sym}^m \mathcal{A} = \binom{e+m}{e} < d$ if $m < h - 1$; there are no

maximal minors. For $A = \langle 1, z_1, \dots, z_e \rangle$ the stable map is surjective, so this threshold cannot be lowered. $\square$

For $e = 2$ this is a genuine three-plane theorem for every $\mathbb{C}[x, y]/(x, y)^h$: the contact length is $\binom{h+1}{2}$ and the failure multiplicity is $\binom{h+1}{3}$. Already $h = 3$ gives a length-six, embedding-dimension-two algebra whose failure divisor has multiplicity four. These are schemes on a full generating Grassmannian, not selected transverse slices. They are distinct from the curvilinear hyperplane schemes and the reduced pair-diagonal ideal.

12.1. A global conductor theorem in arbitrary projective dimension. The preceding algebra is not the algebra of a divisor on a curve. To apply its scheme calculation to global sections, we require a separate conductor theorem. Its range depends on the order h of the fat point, not on its length d .

Theorem 12.2 (Fat-point conductor transport). *Let $e \geq 1$, $h \geq 2$, let $p \in \mathbb{P}^e$, and let Z be the fat point with ideal $\mathcal{I}_p^h$. Suppose $n \geq 2h - 1$. For any generating k -plane $A \subset H^0(Z, \mathcal{O}_Z(n))$, let*

$$W = H^0(\mathbb{P}^e, \mathcal{I}_p^h(n)), \quad U = \text{res}_Z^{-1}(A).$$

For every $m \geq 2$ one has

$$WU^{m-1} = H^0(\mathbb{P}^e, \mathcal{I}_p^h(mn)). \quad (12.4)$$

Restriction induces an isomorphism of coherent cokernels on the entire relative generating Grassmannian, also with p varying,

$$\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow H^0(\mathcal{O}(mn)) \otimes \mathcal{O}) \simeq \text{coker}(\text{Sym}^m \mathcal{A} \rightarrow \pi_* \mathcal{O}_Z(mn)). \quad (12.5)$$

It and all its Fitting ideal equalities commute with arbitrary base change. The bound is sharp uniformly over all generating dimensions k and all $m \geq 2$; no rank-specific sharpness is asserted.

Proof. First fix $p = [1 : 0 : \dots : 0]$ and work on its affine chart with coordinates $z_1, \dots, z_e$. A section of $\mathcal{O}(n)$ is a polynomial of degree at most n , and W has basis the monomials of total degree between h and n . Choose $u \in U$ with constant coefficient u_0 a unit. This is possible over a local parameter base because the augmentation of A and restriction onto A are surjective.

For every integer t with $h \leq t \leq mn$, one can choose $1 \leq j \leq m$ with $jh \leq t \leq jn$. Indeed the integral intervals $[jh, jn]$ have no gaps: $jn + 1 \geq (j + 1)h$ follows from $n \geq 2h - 1$. Every monomial z^α of total degree t can be factored into j monomials of degrees between h and n . Their product, multiplied by u^{m-j} , belongs to WU^{m-1} and has the form

$$u_0^{m-j} z^\alpha + \text{monomials of total degree strictly greater than } t.$$

Its degree is at most mn . Choosing this column for every monomial of degree between h and mn gives a square triangular matrix with units on the diagonal. It is invertible over the coefficient ring. These columns are thus a basis of the whole space on the right of (12.4). The reverse containment is immediate.

Restriction in degrees n and mn is surjective, since those degrees are at least $h - 1$. Its kernels and targets are vector bundles, with monomial bases on the current chart. In the locally split sequence $0 \rightarrow W \rightarrow U \rightarrow A \rightarrow 0$, the kernel on symmetric powers is the image of $W \otimes \text{Sym}^{m-1} U$. Equation (12.4) identifies its multiplication image with the kernel of target restriction. Taking the quotient of this restriction square proves (12.5). The triangular proof works over every coefficient ring once u_0 is a unit. For moving p , choose projective coordinates smoothly locally; restriction and the cokernel isomorphism are intrinsic, so they descend. All the split restriction sequences and surjective maps remain so after arbitrary pullback.

At the boundary $n = 2h - 2$, take $A = \langle 1 \rangle$. Then $U = \langle 1 \rangle + W$, and its quadratic monomials have degrees in $\{0\} \cup [h, 2h - 2] \cup [2h, 4h - 4]$. They omit z_1^{2h-1} , although it lies in the conductor

kernel in degree $2n$. Its nonzero global cokernel class restricts to zero on Z . Thus the cokernel comparison already fails in degree two. $\square$

Corollary 12.3 (Global noncurvilinear primary schemes). *For $e \geq 2$, $h \geq 3$, and $n \geq 2h - 1$, consider all inverse-image series U of generating $(e+1)$ -planes on $Z = (\mathcal{I}_p^h)$. For every $m \geq h - 1$, their global multiplication failure scheme has the exact primary ideal $\mathcal{I}_\Delta^{\binom{e+h-1}{e+1}}$. Every power is primary and there are no embedded associated points on this fixed- p generating parameter space. In degrees $2 \leq m < h - 1$ the failure ideal is zero, so the whole parameter space fails. The same primary assertions hold on the relative parameter space with $p \in \mathbb{P}^e$ varying; the coherent cokernel comparison itself holds after every base change.*

Proof. Apply Theorems 12.1 and 12.2. For moving p there is the following canonical construction. On $P \times P$, $P = \mathbb{P}^e$, let $\mathcal{D}_h$ be the subscheme defined by the h th power of the diagonal ideal, and let p_1, p_2 be the projections. Set

$$\mathcal{B}_h = (p_1)_* \mathcal{O}_{\mathcal{D}_h}, \quad \mathcal{L}_n = (p_1)_* (\mathcal{O}_{\mathcal{D}_h} \otimes p_2^* \mathcal{O}_P(n)).$$

The first is the rank- d jet algebra; the second is an invertible $\mathcal{B}_h$ -module, locally free of rank d over P . Let $\mathcal{M}_n$ be the kernel of $\mathcal{L}_n \rightarrow \mathcal{O}_P(n)$. Its first normal quotient is $\Omega_P^1 \otimes \mathcal{O}_P(n)$. On the relative generating open of $\mathrm{Gr}_P(e+1, \mathcal{L}_n)$, the universal augmentation kernel K therefore has the determinant line map

$$\det K \longrightarrow \det(\Omega_P^1 \otimes \mathcal{O}_P(n)).$$

Its zero divisor is canonical. The map $A \mapsto K$ to $\mathrm{Gr}_P(e, \mathcal{M}_n)$ is an affine-bundle torsor, locally the space of lifts of $\mathcal{O}_P(n)$ in $\mathcal{L}_n/K$. The divisor is the pullback of the relative Schubert divisor for $\mathcal{M}_n \rightarrow \Omega_P^1 \otimes \mathcal{O}_P(n)$. Local splittings of these vector-bundle sequences exhibit the latter as the product with the standard integral Schubert divisor. Since P is integral and the fibres are nonempty, this divisor and its affine-bundle pullback are integral and Cartier. The total generating parameter space is smooth, so the same primary-power argument applies. No primary-decomposition assertion is inferred for arbitrary nonreduced specializations of this parameter space. $\square$

13. PRIMARY DESCENT OVER MULTIPLICITY STRATA

The following descent statement concerns ideals, not just the permutation action on their generic points.

Lemma 13.1 (Etale descent of weighted prime divisors). *Let X be a regular integral Noetherian $\mathbb{C}$ -scheme and $p : \tilde{X} \rightarrow X$ a finite etale Galois cover, with group G . Let $\{D_\alpha\}$ be a finite G -stable collection of distinct integral effective Cartier divisors on $\tilde{X}$. Work componentwise if the cover is disconnected. Suppose the positive integers b_α are constant on G -orbits. For an orbit O , there is a unique integral effective Cartier divisor D_O on X whose ideal satisfies*

$$p^* \mathcal{I}_{D_O} = \prod_{\alpha \in O} \mathcal{I}_{D_\alpha} = \bigcap_{\alpha \in O} \mathcal{I}_{D_\alpha}.$$

If a descended ideal $\mathcal{J}$ pulls back to $\prod_\alpha \mathcal{I}_{D_\alpha}^{b_\alpha}$, then its complete primary decomposition is

$$\mathcal{J} = \prod_O \mathcal{I}_{D_O}^{b_O} = \bigcap_O \mathcal{I}_{D_O}^{b_O}. \quad (13.1)$$

There are no embedded associated points, and the formula holds at intersections as well as generically. The descended prime divisor need not be smooth or geometrically unbranched.

Proof. In each regular local ring upstairs, equations for distinct prime Cartier divisors are distinct prime factors. Unique factorization gives product equal to intersection, with or without positive powers. The reduced orbit union has an invariant invertible ideal subsheaf of $\mathcal{O}_{\tilde{X}}$ with its canonical descent data. Faithfully flat descent gives an invertible ideal subsheaf of $\mathcal{O}_X$. It defines an effective Cartier divisor. Its support is the image of any one member of the orbit, hence is irreducible, since the map is finite. Reducedness descends faithfully flat, so this divisor is integral.

The same factorization upstairs, and flatness for finite intersections, identify the pullbacks of all three ideals in (13.1). Faithfulness makes them equal downstairs. Locally downstairs each $\mathcal{I}_{D_O}$ is generated by a prime element of a regular local ring; its b_O th power is primary. Their intersection is irredundant after discarding components absent from the chart, so it is the complete primary decomposition and has no embedded primes. No normal-crossing hypothesis was used. An orbit can have several members through a point upstairs, giving several etale branches of the integral descended divisor. This explains the final distinction. $\square$

Apply the lemma to the relative generating pencil Grassmannian over a multiplicity stratum. Ordering the distinct support points gives a finite etale cover with permutations among equal multiplicities. On it the divisors T_i, E_{ij} are the relative rank-one Schubert divisors. Their ideals are prime Cartier, and their weights are $\binom{d_i}{2}, d_i d_j$. Formula (13.1) supplies the whole descended primary decomposition. In particular it remains valid at a braid intersection where several values agree. The fixed labelled divisors are smooth; the descended orbit divisors are asserted integral, not automatically smooth or unbranched.

This argument applies to the earlier pencil calculation. For the contact-hyperplane family, Theorem 5.2 gives a scheme isomorphism with the quotient incidence and does not require a conjectural primary descent rule.

14. CONTACT PENCILS: EQUATIONS, PRIMARY COMPONENTS, AND INFINITESIMAL LAYERS

Fix $d \geq 4$ and $n \geq 2d - 2$. In this section the divisor D is fixed, although it need not be reduced. Set $G_D^\circ = \text{Gr}(2, E_{n,D})^\circ$. It is a smooth irreducible variety of dimension $2d - 4$. The conductor construction embeds it as the open generating locus in the Grassmannian of subseries containing W_D . Let $\mathcal{J}_m$ be the zeroth Fitting ideal of $\text{coker}(\text{Sym}^m \mathcal{U} \rightarrow V_{mn})$ on this variety.

Trivialize $\mathcal{O}_D(n)$, and put $R = H^0(\mathcal{O}_D)$. There is a smooth surjective frame map

$$\mathcal{F}_D = \{(a, v) : a \in R^\times, v \in R \setminus \mathbb{C}\} \longrightarrow G_D^\circ, \quad (a, v) \longmapsto \langle a, av \rangle. \quad (14.1)$$

It is the open part of the frame bundle where the first basis vector is a unit. Its relative dimension is four. All calculations on $\mathcal{F}_D$ below are therefore exact faithfully flat local calculations for G_D° , not transverse slices inferred from pointwise ranks.

Lemma 14.1 (The power-basis presentation). *On $\mathcal{F}_D$ the multiplication cokernel is presented by*

$$K_m(v) = [1, v, v^2, \dots, v^m] : \mathcal{O}^{m+1} \longrightarrow R \otimes \mathcal{O}. \quad (14.2)$$

For $m \geq d - 1$, every later column is an $\mathcal{O}$ -linear combination of the first d columns, and

$$\mathcal{J}_m \mathcal{O}_{\mathcal{F}_D} = (\Delta_D(v)), \quad \Delta_D(v) = \det[1, v, \dots, v^{d-1}]. \quad (14.3)$$

For $2 \leq m < d - 1$, $\mathcal{J}_m = 0$.

Proof. By Theorem 2.2, it suffices to multiply sections in the pencil $\langle a, av \rangle$ on D . Its image is generated by $a^m v^j$ for $0 \leq j \leq m$. Multiplication by a^{-m} is an invertible target operation and gives (14.2). Multiplication by v is an endomorphism of the rank- d free module $R \otimes \mathcal{O}$. Its monic characteristic polynomial annihilates it over the coefficient ring, by the adjugate proof of Cayley–Hamilton. Thus v^d and every higher power lie in the span of the first d powers, with regular coefficients. Column operations reduce the presentation to $[K_{d-1}(v), 0]$. This proves

(14.3), as an identity of ideals and not merely of radicals. If $m + 1 < d$, all d -minors are zero. Faithful flatness descends this assertion. $\square$

Write $D = \sum_{i=1}^s d_i p_i$, with distinct p_i . Choose local parameters z_i and the Chinese-remainder identification

$$R = \prod_{i=1}^s \mathbb{C}[z_i]/(z_i^{d_i}), \quad v_i = \lambda_i + \sum_{r=1}^{d_i-1} c_{ir} z_i^r.$$

All λ_i, c_{ir} are independent affine coordinates before removing the scalar locus. For $d_i \geq 2$ define T_i by failure of A to separate $2p_i$; for $i < j$ define E_{ij} by failure to separate $p_i + p_j$. On the frame space they have equations $c_{i1} = 0$ and $\lambda_j - \lambda_i = 0$, respectively.

Theorem 14.2 (Full fixed-contact classification). *For every $m \geq d - 1$, in the ordered jet basis $1, z_i, \dots, z_i^{d_i-1}$,*

$$\Delta_D(v) = \prod_{i:d_i \geq 2} c_{i1}^{\binom{d_i}{2}} \prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}. \quad (14.4)$$

The complete primary decomposition on G_D° is

$$\mathcal{J}_m = \bigcap_{i:d_i \geq 2} \mathcal{I}_{T_i}^{\binom{d_i}{2}} \cap \bigcap_{i < j} \mathcal{I}_{E_{ij}}^{d_i d_j}. \quad (14.5)$$

All displayed divisors are nonempty, smooth, irreducible, and distinct. They are precisely the associated subvarieties; no embedded associated points occur. The failure divisor has dimension $2d - 5$ and divisor class $\binom{d}{2}$ times the restricted Pluecker class.

Every multiplication corank, for every $m \geq 2$, is determined as follows. Put $e_i = \text{ord}_{z_i}(v_i - \lambda_i)$, with $e_i = d_i$ for a constant v_i , and put $\ell_i = \lceil d_i/e_i \rceil$. For each distinct value λ among the λ_i , set $L_\lambda = \max_{i:\lambda_i=\lambda} \ell_i$, and let $s(v) = \sum_\lambda L_\lambda$. Then

$$\text{corank}(\text{Sym}^m U_A \rightarrow V_{mn}) = d - \min\{m + 1, s(v)\}. \quad (14.6)$$

On the frame space every higher Fitting ideal is, exactly, the corresponding minor ideal of (14.2).

Proof. For a polynomial P of degree at most $d - 1$, Taylor expansion gives

$$P(v_i(z_i)) = \sum_{r=0}^{d_i-1} \frac{P^{(r)}(\lambda_i)}{r!} (v_i(z_i) - \lambda_i)^r \pmod{z_i^{d_i}}.$$

The linear change from the divided derivatives of P at λ_i to the coefficients of $P(v_i(z_i))$ is triangular, with diagonal $1, c_{i1}, \dots, c_{i1}^{d_i-1}$. Its determinant is $c_{i1}^{\binom{d_i}{2}}$, including when this coefficient is zero. The determinant of the divided-derivative evaluation map

$$P \mapsto \left(\frac{P^{(r)}(\lambda_i)}{r!} \right)_{i, 0 \leq r < d_i}$$

in the basis $1, T, \dots, T^{d-1}$ is the confluent Vandermonde $\prod_{i < j} (\lambda_j - \lambda_i)^{d_i d_j}$. Here is an explicit normalization of that identity. Start with d distinct variables, cluster d_i of them near λ_i , and in each cluster replace successive evaluation rows by divided-difference rows. The ordinary Vandermonde determinant is divided by the product of the differences *within* each cluster. In the cluster limit the rows are the divided derivatives above, while the remaining cross-cluster factors tend to the displayed product. This proves the identity on a dense open set and hence as a polynomial identity, including all collisions. Combining the two determinants proves (14.4) without inverting any of its factors.

For any length-two $Z \leq D$, restriction $E_{n,D} \rightarrow E_{n,Z}$ is surjective. The locus in $\text{Gr}(2, E_{n,D})$ where its restriction to A has rank at most one is the corresponding irreducible Schubert divisor.

Its rank-zero locus is excluded by generation on D . Its rank-one locus is smooth: choose a basis of A with one nonzero image, and vary the second basis vector in a complementary target direction; the determinant then has a nonzero linear derivative. For irreducibility one may also mark the image line in $E_{n,Z}$; the resulting incidence is a Grassmannian bundle over $\mathbb{P}(E_{n,Z})$, and projects birationally onto the rank-one locus. The generating open is nonempty for each divisor when $d \geq 4$. On frames this follows directly by choosing $a = 1$ and a nonscalar v with the single prescribed linear equality. The same coordinates show that the divisors are distinct.

On each affine frame chart the coordinate ring is a localization of a polynomial ring. The distinct linear factors in (14.4) generate distinct prime ideals there. Unique factorization gives $(\prod f_\alpha^{b_\alpha}) = \bigcap (f_\alpha^{b_\alpha})$; each ideal $(f_\alpha^{b_\alpha})$ is primary. These identities descend by faithful flatness to (14.5). Equivalently, at each point of the smooth base the local ring is regular, each displayed divisor is smooth, and its powers are primary. This also proves absence of embedded primes. The dimension follows since this is a nonzero Cartier divisor. Each component is a Pluecker hyperplane section, and

$$\sum_i \binom{d_i}{2} + \sum_{i < j} d_i d_j = \binom{d}{2},$$

which proves the class formula.

Finally, in $\mathbb{C}[z_i]/(z_i^{d_i})$ the element $v_i - \lambda_i$ has nilpotency index ℓ_i . Its minimal polynomial is $(T - \lambda_i)^{\ell_i}$. In the product algebra the minimal polynomial is therefore

$$P_v(T) = \prod_\lambda (T - \lambda)^{L_\lambda}. \quad (14.7)$$

No polynomial of smaller degree can annihilate v , and division by this monic polynomial shows that $1, v, \dots, v^{s(v)-1}$ form a basis of $\mathbb{C}[v]$. Consequently the rank of $K_m(v)$ is $\min\{m+1, s(v)\}$. The cokernel isomorphism of Theorem 2.2 gives (14.6) and the assertion about Fitting ideals. $\square$

Theorem 14.3 (Exact infinitesimal layers and singularities). *Let $m \geq d-1$. At a point of a smooth frame chart discard the nonvanishing factors of (14.4). Write the remaining distinct linear factors as f_α , with multiplicities b_α , and put*

$$F = \prod_\alpha f_\alpha^{b_\alpha}, \quad P = \prod_\alpha f_\alpha.$$

In the regular local ring B of that chart the failure ring is $B/(F)$ and its nilradical is $\mathcal{N} = (P)/(F)$. Its exact nilpotency index is $\max_\alpha b_\alpha$, with index one for a reduced local ring. For every $q \geq 1$, there is an isomorphism of B -modules, with indicated generator P^q ,

$$\mathcal{N}^q / \mathcal{N}^{q+1} \simeq B / \left(P, \prod_\alpha f_\alpha^{(b_\alpha - q)_+} \right). \quad (14.8)$$

The completed local equation is the same weighted linear arrangement, with unused frame and higher-jet coordinates as smooth parameters. Its tangent cone is defined by the homogeneous product of the active linear factors with these same multiplicities. Its singular scheme on this chart has ideal $(F, \partial F)$, and its singular support is the union of the components with $b_\alpha \geq 2$ and the pairwise intersections of distinct components. These descriptions descend smoothly to G_D° .

Proof. The local ring in question is a localization of a polynomial ring in the unit coordinates of a and the independent jet coordinates of v . The factors f_α are prime, distinct, and vanish at the selected point. Unique factorization shows that $P^q \in (F)$ if and only if $q \geq b_\alpha$ for every α . This proves the nilpotency assertion, including at multiple intersections.

The homomorphism $B \rightarrow \mathcal{N}^q/\mathcal{N}^{q+1}$ sending 1 to the class of P^q is onto. Its kernel is

$$((P^{q+1}, F) : P^q) = (P) + (F : P^q) = \left(P, \prod_{\alpha} f_{\alpha}^{(b_{\alpha}-q)+} \right).$$

For the middle equality, if $rP^q = sP^{q+1} + tF$, subtract sP from r ; cancellation gives membership in $(F : P^q)$. The last colon is computed by prime-factor exponents. This proves (14.8). Globally the chosen generator carries the corresponding line-bundle trivialization; the displayed isomorphism is a local one and makes no claim of a preferred global generator.

At the chosen point every active factor is linear in centered coordinates. Every discarded factor is a unit, so the completed ideal is exactly the displayed product, not a truncation of it. The initial ideal is generated by the same homogeneous product, which proves the tangent-cone statement. The hypersurface cotangent presentation gives $(F, \partial F)$. At a point on exactly one component of multiplicity one the differential is a unit times $df_{\alpha} \neq 0$. At every point on a multiple component or on two distinct components it is zero. This proves the singular support assertion. Smooth faithfully flat pullback preserves these local statements; in particular the four frame parameters do not alter the nilpotency index or the singular-support criterion. $\square$

For illustration, length four already gives subseries of codimension two, in every V_n with $n \geq 6$. For every $m \geq 3$ their fixed-contact primary equations are as follows. Put $\delta_{ij} = \lambda_j - \lambda_i$; only first-jet coefficients of multiple points are displayed.

Multiplicities of D	Exact equation on frames	Largest local nilpotency index
$(1, 1, 1, 1)$	$\prod_{i < j} \delta_{ij}$	1
$(2, 1, 1)$	$c_{11} \delta_{12}^2 \delta_{13}^2 \delta_{23}$	2
$(2, 2)$	$c_{11} c_{21} \delta_{12}^4$	4
$(3, 1)$	$c_{11}^3 \delta_{12}^3$	3
(4)	c_{11}^6	6

The reduced arrangement need not have normal crossings: three coincident values give three braid hyperplanes in a two-dimensional normal space. The equation, rather than a normal-crossing label, records that singularity. Formula (14.6) also distinguishes points with the same support of failure but different higher-jet vanishing orders.

15. MOVING CONTACTS AND THE NORMALIZATION OF THE FAILURE DIVISOR

Let $X_d^{\circ} = X_{d,2}^{\circ}$ be the relative generating Grassmannian in Theorem 2.2. It is smooth and irreducible of dimension $3d - 4$. The fixed-divisor schemes of the preceding section are its scheme-theoretic fibres, not its irreducible components. For $m \geq d - 1$ denote its failure scheme by Y_m .

15.1. Components over every multiplicity stratum. For a partition $\mathbf{d} = (d_1, \dots, d_s)$ of d , let $B_{\mathbf{d}} \subset \text{Sym}^d \mathbb{P}^1$ be the locally closed stratum of divisors with exactly this multiplicity partition, and restrict X_d° to it. Let M be the set of distinct multiplicities, and let $r_e = \#\{i : d_i = e\}$ for $e \in M$.

Corollary 15.1 (Complete component descent on multiplicity strata). *For $m \geq d - 1$, the failure scheme over $B_{\mathbf{d}}$ has*

$$\#\{e \in M : e \geq 2\} + \binom{\#M}{2} + \#\{e \in M : r_e \geq 2\} \quad (15.1)$$

irreducible components, all of dimension $s + 2d - 5$, and no embedded associated points. A ramification component is indexed by each $e \geq 2$ in M and has primary multiplicity $\binom{e}{2}$. An

identification component is indexed by each unordered pair of multiplicities e, f that occurs at distinct support points and has primary multiplicity ef . The ideal of the scheme is the intersection of these prime-divisor ideals raised to the stated powers. On the labelled étale cover its full local equations and nilradical layers are exactly those of Theorem 14.3.

Proof. The ordered configuration space of s distinct points of $\mathbb{P}^1$ is an integral finite étale cover of B_d , with deck group $\prod_{e \in M} \mathfrak{S}_{r_e}$. The relative first-jet and two-point restriction maps are surjective over this cover. Consequently each labelled T_i and E_{ij} is an irreducible relative rank-one Schubert divisor. The local factorization (14.4) holds with the support coordinates free, so there are no further components. The deck group permutes T_i exactly according to the value d_i , and permutes E_{ij} exactly according to the unordered pair $\{d_i, d_j\}$. The images of the orbit unions are precisely the irreducible components downstairs: each is the image of an integral labelled component, and two such images agree exactly for components in the same orbit. This gives (15.1).

Lemma 13.1 applies to this regular ambient space and its invariant weighted divisors, including all branch intersections. It proves that each component downstairs is a prime Cartier divisor; irreducibility here does not assert smoothness or geometric unbranchedness. Pulling its reduced ideal back to the labelled cover gives the reduced product of its labelled branch equations. Their multiplicity is constant on an orbit, so the primary ideal identity in (14.5) descends faithfully flatly. This also proves absence of embedded points and gives the stated local layers. The base stratum has dimension s , giving the dimension formula. A descended component can have intersecting local branches; smoothness of each labelled divisor is not a claim that every descended component is smooth. $\square$

Theorem 15.2 (An integral failure divisor with smooth normalization). *For $d \geq 4$, $n \geq 2d - 2$, and $m \geq d - 1$, the schemes Y_m are all the same integral Cartier divisor Y of dimension $3d - 5$. In particular Y has no embedded associated points. Its normalization is*

$$\tilde{Y} = \{(D, A, Z) : (D, A) \in X_d^\circ, Z \leq D, \deg Z = 2, \text{rank}(A \rightarrow H^0(\mathcal{O}_Z(n))) = 1\}. \quad (15.2)$$

The forgetful map $\tilde{Y} \rightarrow Y$ is finite and birational, and $\tilde{Y}$ is smooth and irreducible.

More explicitly, on an affine support chart write $D = (g(t))$ with g monic of degree d , and on a frame chart put $v = b/a \in \mathbb{C}[t]/(g)$, represented by a polynomial of degree less than d . The failure equation is

$$\Delta(g, v) = \det[1, v, \dots, v^{d-1}]_{\mathbb{C}[t]/(g)} = 0. \quad (15.3)$$

The normalization on this chart is parametrized by

$$g = hq, \quad v = \lambda + hw, \quad (15.4)$$

where h, q are monic of degrees $2, d - 2$, respectively, $\lambda \in \mathbb{C}$, and $\deg w \leq d - 3$. The scalar-pencil locus is omitted; the unit a is a free smooth parameter.

Proof. Theorem 2.2 applies on the whole relative space. The proof of Lemma 14.1 applies over the polynomial coefficient ring of a varying monic g , not only on its fibres. Thus every Y_m has the identical ideal (15.3) on a smooth covering by frame charts. This determinant is not identically zero: $v = t$ makes its matrix the identity for every g .

The space of pairs $Z \leq D$ with lengths $2, d$ is $\text{Sym}^2 \mathbb{P}^1 \times \text{Sym}^{d-2} \mathbb{P}^1$; the map sends (Z, Q) to $D = Z + Q$. It is finite. Indeed it is proper, and a fixed effective divisor has only finitely many degree-two subdivisors, obtained by choosing multiplicities at its finitely many support points. Restriction $E_{n,D} \rightarrow E_{n,Z}$ is a surjection of vector bundles. The rank-at-most-one condition on a pencil is a relative Schubert divisor; its rank-zero part is excluded by the generating hypothesis. This makes (15.2) a closed subscheme of the finite base change to X_d° , so its projection is finite.

Smoothness and irreducibility can be checked explicitly. A divisor with finite support is contained in an affine coordinate chart. On that chart Z is defined by a unique monic quadratic

h , and $Z \leq D$ means $g = hq$. Since a is a unit, rank one on Z is equivalent to $v \bmod h$ being a scalar λ . Division by the monic h gives uniquely $v = \lambda + hw$, with the stated degree bound. Thus the framed incidence is an open part of affine space with coordinates the coefficients of h, q, w , the scalar λ , and the unit a . This proves smoothness. Globally irreducibility also follows from the relative rank-one Schubert description over the integral pair-divisor base; each fibre is irreducible, and its generating open is nonempty. These charts describe the same incidence, so no extra component appears at infinity.

An element v fails to generate the length- d algebra $R = H^0(\mathcal{O}_D)$ precisely when it either takes the same value at two distinct support points or has zero first derivative at a multiple support point. This follows at once from the minimal polynomial (14.7): its degree is d if and only if the support values are distinct and each local $\ell_i = d_i$. These two failure mechanisms are precisely the existence of a length-two $Z \leq D$ on which v is constant. Therefore the image of the finite incidence is the whole support of (15.3). That support is irreducible.

To determine its multiplicity, work over divisors with distinct support, label their roots after an étale base change, and use evaluation coordinates for v . The fixed-fibre formula becomes a product of the distinct differences $\lambda_j - \lambda_i$. At a point with exactly one equal pair the determinant has a nonzero linear derivative. Hence its multiplicity along its unique irreducible support is one. Since the ambient space is smooth, a Cartier hypersurface has no embedded associated points; a generically reduced such hypersurface is reduced. One may see the last assertions locally from unique factorization in the regular local ring, or from the Cohen–Macaulay and $R_0 + S_1$ criterion [24, Tag 031R]. Thus Y is integral.

Over the dense open subset with reduced D and precisely one equal pair there is exactly one possible Z . The finite projection is therefore birational in characteristic zero. Its source is smooth and hence normal, so it is the normalization. The affine parametrization is exactly (15.4), as already proved. The dimension of the divisor is $3d - 5$. $\square$

Proposition 15.3 (Exact singular-support test across collisions). *At a closed point $(D, A) \in Y$, call a length-two $Z \leq D$ scalar if A has rank one on Z . The point is smooth on Y if and only if there is precisely one scalar Z and the following linear system has no nonzero solution. In the associated factorization (15.4), the unknown is $r(t)$ of degree less than two, and the system is*

$$h \mid qr, \quad wr \bmod h \text{ is constant.} \quad (15.5)$$

In particular, this criterion is effective at nonreduced D and is not merely the distinct-support pair-count criterion. The full singular scheme on a frame chart has ideal

$$(\Delta, \partial\Delta/\partial g_0, \dots, \partial\Delta/\partial g_{d-1}, \partial\Delta/\partial v_0, \dots, \partial\Delta/\partial v_{d-1}). \quad (15.6)$$

Proof. If the normalization has more than one point above (D, A) , the local ring of Y cannot be regular. Suppose there is just one. We first justify the differential criterion for a finite normalization $\nu : N \rightarrow Y$ with N smooth. A regular local ring is normal, so a smooth point of Y has an isomorphic normalization near it. Conversely let $A_0 \subset B_0$ be the local finite extension at a point with a unique preimage and common residue field $\mathbb{C}$. Injectivity of $d\nu$ means

$$\mathfrak{m}_{B_0}/(\mathfrak{m}_{B_0}^2 + \mathfrak{m}_{A_0}B_0) = 0.$$

Nakayama’s lemma first gives $\mathfrak{m}_{B_0} = \mathfrak{m}_{A_0}B_0$. It then gives, for the finite A_0 -module B_0 , that 1 generates B_0 , since the quotient by $\mathfrak{m}_{A_0}$ is $\mathbb{C}$. The injective map $A_0 \rightarrow B_0$ is consequently surjective. Thus $A_0 = B_0$ is regular. Local finiteness here follows by localizing the finite morphism; uniqueness of the preimage makes the resulting semilocal ring local.

In the affine normalization coordinates, a tangent vector with $\dot{g} = \dot{v} = 0$ satisfies

$$0 = q\dot{h} + h\dot{q}, \quad 0 = \dot{\lambda} + w\dot{h} + h\dot{w}.$$

Set $r = \dot{h}$, so $\deg r < 2$ because h is monic. The first equation is soluble exactly when $h \mid qr$, and then determines $\dot{q} = -qr/h$. The second is soluble exactly when the remainder of wr modulo h is constant; it determines $\dot{\lambda}$ as the negative of that constant and determines $\dot{w}$ by division. The resulting degrees satisfy the required bounds. If $r = 0$ all four variations are zero. Thus the differential has nonzero kernel exactly when (15.5) has a nonzero solution. The unit frame parameter has zero variation in a kernel and creates no additional one. Smooth frame descent proves the criterion on Y . Finally (15.6) is the hypersurface Jacobian presentation in its smooth affine ambient space. $\square$

Remark 15.4 (A smooth total point with a triple primary fibre). Take $d = 4$, $g = t^3(t - 1)$, and $A = \langle 1, t^2 \rangle$ on $D = (g)$. The only scalar length-two subdivisor is $h = t^2$. In (15.4) one has $q = t(t - 1)$ and $w = 1$. For $r = r_0 + r_1 t$, divisibility $h \mid qr$ forces $r_0 = 0$, while constancy of $wr \bmod h$ forces $r_1 = 0$. Hence the total divisor is smooth here. On the fixed-divisor fibre, the two support values are distinct and (14.4) is a unit times c_{11}^3 . Its exact local nilpotency index is three. Thus even a smooth point of the integral total divisor can have a nonreduced primary fibre.

Proposition 15.5 (Discriminants and specialization). *Let $\chi_v(T)$ be the characteristic polynomial of multiplication by v in $\mathbb{C}[t]/(g)$. With the usual monic polynomial normalization of discriminants,*

$$\text{disc}(\chi_v) = \text{disc}(g) \Delta(g, v)^2. \quad (15.7)$$

This is a polynomial identity on the entire coefficient space. The equation $\Delta = 0$, not $\text{disc}(\chi_v) = 0$, defines the failure divisor, including when g has multiple roots. Its fibre over each fixed D is exactly the primary scheme in (14.5).

Proof. When g has distinct roots $t_1, \dots, t_d$, the evaluation matrix changes the basis $1, t, \dots, t^{d-1}$ to point values. Therefore

$$\Delta(g, v) = \frac{\prod_{i < j} (v(t_j) - v(t_i))}{\prod_{i < j} (t_j - t_i)}.$$

Squaring gives (15.7), because $\chi_v(T) = \prod_i (T - v(t_i))$ on this open set. Both sides are polynomial in the coefficients, so the identity holds everywhere. For $v = t$ the determinant is one even when g is nonreduced; this also shows why the factor $\text{disc}(g)$ must not be included in the failure equation. The scheme-theoretic fibre statement follows from Fitting base change in Theorem 2.2, or by specializing the polynomial determinant and applying (14.4). $\square$

Proof of Theorem 1.1. The fixed-contact assertions are Theorems 14.2 and 14.3. The moving-contact assertions are Theorem 15.2 and Proposition 15.3. All four concern the original multiplication map, by the coherent cokernel isomorphism of Theorem 2.2. $\square$

16. PENCIL COLLISIONS AND COMPARISON OF LOCAL TYPES

The normalization criterion above separates phenomena which a set-theoretic discriminant calculation conflates. In the affine normalization write $g = hq$ and $v = \lambda + hw$, with h monic of degree two. The kernel test is

$$h \mid qr, \quad wr \bmod h \text{ constant}, \quad \deg r < 2.$$

The following three cases give concrete local distinctions.

If g has distinct roots, the normalization is unramified; a unique equal value pair gives a smooth point, and two disjoint equal pairs give an ordinary double crossing. Three equal values give the three braid factors $\delta_{12}\delta_{13}(\delta_{13} - \delta_{12})$, not a normal-crossing divisor in a two-dimensional normal space. These are exact equations because root labelling is étale here.

For $g = t^3(t - 1)$ and $v = t^2$, the scalar subdivisor $h = t^2$ is unique. One has $q = t(t - 1)$ and $w = 1$. Divisibility forces the constant term of r to vanish and the second condition forces its linear term to vanish. The total pencil divisor is smooth, although its fixed- D fibre has primary equation a unit times c_{11}^3 .

For $g = t^4$ and $v = t^2$, again $h = t^2$ is the unique scalar subdivisor. Now $q = t^2$ and $w = 1$. The constant $r = 1$ satisfies the kernel test. Thus the total pencil divisor is singular and its normalization is ramified at the unique preimage. This last example is asserted to have a unique normalization point, not an unproved analytic cusp normal form. In contrast, the moving contact-hyperplane scheme is smooth at all these divisor collisions by Theorem 5.4. Its fixed fourfold-contact transverse algebra is G_4 , with length six and nilpotency index five, rather than the pencil's sixth power of a prime divisor.

17. RELATION TO EARLIER CONSTRUCTIONS

The determinant $\det[1, v, \dots, v^{d-1}]$ is a classical local index form. Arpin, Bozlee, Herr and Smith construct schemes of monogenerators and polygenerators for finite locally free algebras [29]. Their Proposition 3.6 and Definition 3.12 identify the index-form open and its closed complement; Proposition 3.14 gives the polygenerator open by minors. Thus neither the generator functor nor its index determinant is claimed as new here. Our use concerns the coherent cokernel of a global section-multiplication map and, for generating hyperplanes, its exact closed incidence with a rank-two quotient. Lemma 5.1 and Theorem 5.2 prove that closed-scheme identification on nonreduced bases, rather than deriving it from agreement of generator opens.

The algebra G_r is the classical two-row Grassmannian cohomology ring; [31, Theorem 2.7] provides the rectangular Schur basis in a more general quotient. Its role here is an exact transverse algebra of multiplication failure, determined by a rank-two quotient of $\mathbb{C}[z]/(z^r)$. The explicit residual equations (5.7) verify this identification without a tangent-cone approximation. The diagonal-ideal and all-powers identities in the three-plane result are precisely [30, Corollary 3.8.3]; the argument here supplies the finite total-degree presentation and conductor transport, not a new proof of that theorem.

Normal generation in sufficiently positive degree on a curve is classical [14]. The curve conductor theorem uses it only for $L(-D)$, and then proves the precise quotient comparison by filling the double-vanishing space and its first normal layer. On $\mathbb{P}^1$ the evaluation-kernel argument gives the sharper rank-dependent bound and its boundary example.

For pencils, confluent Vandermonde factorization and the discriminant identity are classical algebra. The moving normalization is the explicit scalar-pair incidence of that index-form divisor. Its finite map and differential test are proved in Section 15; the normalization is not offered as a construction independent of index-form geometry. The contact-hyperplane incidence is different: it is the failure scheme itself, not just its normalization, and its total space is smooth even at collisions.

Subalgebra varieties, quotient flags, and the additional comparisons. Iovanov and Sistko [32, Theorem 0.1] classify maximal subalgebras of finite-dimensional algebras over fields under their Schur hypothesis. In the commutative algebraically closed case, the alternatives include identifying two residue factors and imposing a first-order condition at one factor. These closed-point mechanisms are classical and are not claimed here as new. Sistko [33, Definition 3.1, Theorems 1.1 and 4.1] constructs subalgebra varieties and computes the maximal-subalgebra vanishing ideal for basic algebras. The distinction needed here is between such a variety and the full relative quadratic-equation functor: for $\mathbb{C}[z]/(z^r)$ the latter already has the transverse algebra G_r , of length $\binom{r}{2}$. Agreement on the reduced variety does not identify this thickening.

The Hilbert functor used in Section 5 is the standard functor of locally free quotient algebras; its representability is not a contribution of this article. Likewise, nested effective divisors on a curve are classically expressed by a divisor and its residual divisor. What is established here is the identification of these functors with specified multiplication Fitting schemes, the canonical cokernel line (5.2), and the global-section transports. These are separate assertions from the classical classification of geometric hyperplanes.

The faithful-action viewpoint is also classical; compare [33, Proposition 3.4] for subalgebras of matrix algebras. Our conductor flags record the *specified inclusion* into B , the action on B/E , and its exact-rank scheme structure. A conductor rank bound alone is not the flag equivalence, nor does either imply unrestricted base change across an action-rank jump. The one-variable codimension-two classification is already given in [34, Theorem 39]. We use it to locate the fibres of Example 6.5, not as a new classification of polynomial subalgebras. The quotient-tower statement concerns arbitrary finite B and its relative exact-rank functors.

In [29, Example 4.16], the two-generator jet calculation has curvilinear target $k[\varepsilon]/\varepsilon^{m+1}$. Our target in (12.1) has embedding dimension e . The symmetric-power determinant identity in the proof is classical linear algebra. Its role is to determine the full primary power of the tangent determinant, uniformly in e, h , and then identify that power with global section multiplication on $\mathbb{P}^e$ using Theorem 12.2. Merely knowing the polygenerator open would not determine this exponent or the global degree range. These comparisons identify the inputs and additional claims; they are not a certificate of exhaustive priority against all literature.

Codimension, embedded incidences, and regularity. The elementary dimension bound for an increasing sequence of subspaces is not a scheme-theoretic stabilization theorem. The proof of Theorem 4.1 retains the compressed commutator relation over the coefficient ring before using polarized Cayley–Hamilton. The polygenerator minors of [29] remain the classical ambient construction. The assertions added here are the codimension-sensitive degree bound, the full codimension-two matrix, and its conductor incidence on nonreduced bases. We do not claim an exhaustive priority comparison for every possible length bound.

The binary cubic for a rank-three algebra is the explicit square condition for a line to lift to a subalgebra. No independent novelty claim is made for that classical cubic-algebra calculation. Its role in Proposition 7.2 is to describe an exceptional fibre of the same incidence which transfers the embedded local rings in Theorem 9.1. The contact algebra G_h and its Schur basis are classical inputs; the mixed equations bv, b^2u , their displayed primary intersection, and their realization as local rings of full multiplication failure are proved separately.

Sidman’s Theorems 1.3 and 1.8 in [35] supply the relationship between regularity and saturation, and the ordinary-square estimate $\text{reg}(I^2) \leq 2 \text{reg } I$ for a finite projective scheme. Sidman also records the earlier power bounds of Chandler and of Geramita–Gimigliano–Pitteloud. None of these regularity inequalities is claimed as new. Their use here is the normal-layer identity (10.8) and the resulting isomorphism of actual multiplication cokernels, including relative families. This separation is particularly important: bounding the regularity of an ideal is not by itself a theorem identifying the Fitting scheme of a prescribed incomplete linear series.

Ballico’s failure-locus work [4, 5] is a necessary nearest-source comparison. The accessible statements of [5, Theorem 0.2, Proposition 2.2, Remark 2.3] concern failure on finite linear sections of a fixed completely embedded variety. The complete theorem pages of [4] have not been obtained in this revision, so their exact relation to the present scheme statements remains unverified. We make no inference of non-anticipation from that access limitation. The results here are submitted with complete proofs and explicit classical inputs; priority of the resulting package is a separate question for source comparison and editorial assessment.

17.1. Generators, algebra length, and the boundary classification. The scheme-theoretic study of generators is an established input. Arpin–Bozlee–Herr–Smith [29] construct the index-form non-monogenerator scheme (Definition 3.12) and describe the polygenerator open by maximal minors (Proposition 3.14); the numbering here refers to the identified preprint version. Thus neither representability, index forms, nor the use of maximal minors is a novelty claim of this paper. In their construction Cayley–Hamilton bounds powers of each individual generator. The additional tasks here are the codimension-sensitive equality of image modules, descent to the generating Grassmannian, the conductor incidence, and the complete primary decompositions of the full boundary in the stated classes.

The field-level length inequality is separated from the relative assertion in Section 4.1. We do not use a special numerical estimate from the algebra-length literature as an unproved premise. The elementary dimension argument and the sharpness family are proved in the text. The coefficient-ring proof supplies the Fitting-ideal consequence needed when the base is nonreduced.

Theorems 8.1–8.3 identify the present boundary contribution precisely: a full multiplication ideal, all its associated strata for Hilbert function $(1, 2, 2)$, their primary representatives, and their conductor position. They do not assert a primary classification for every finite algebra. The higher-contact and complete contact-hyperplane results remain separate theorems with their original scope. The discussion of [4] in Section 17 remains subject to the same source-access limitation; no exhaustive priority comparison with that article is asserted on the basis of its title or bibliographic record.

17.2. Logical dependencies. The finite-algebra core is self-contained. It does not take a theorem from a manuscript labelled A1 as an unstated premise. The evaluation and conductor comparisons proved here are the interfaces through which a global linear-series problem becomes a finite-algebra problem. The outputs available to further work are canonical multiplication cokernels and their Fitting ideals, exact-action conductor flags, and the primary boundary classification. A downstream application must verify the stated generating and regularity hypotheses and retain the scheme structure; the determinant support alone loses the embedded strata. The historical polar, residual and statistical sections are not prerequisites for this core. This logical dependency statement does not certify that any particular version of a later manuscript already invokes these results.

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